

E-tab 4

Service Manual

R2.0

THINKWARE SYSTEMS CORPORATION

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 - **How to repair**
- **Frequently happening problem**

HW spec

Main components

CPU - SC54412ACD-A040 [SAMSUNG]

ARM Cortex-A9 Quad core, 1.4GHz

LPDDR2 1GB, 400MHz (POP)

Capacity - KLMAG4FE3B-A001 [SAMSUNG]

MoviNAND, 16GB

Wi-Fi / BT Combo - WM-BAN-BM-07-S [USI]

802.11a/b/g/n, BT Class 4.0+ HS

GPS – UBX-G6010 [Ublox]

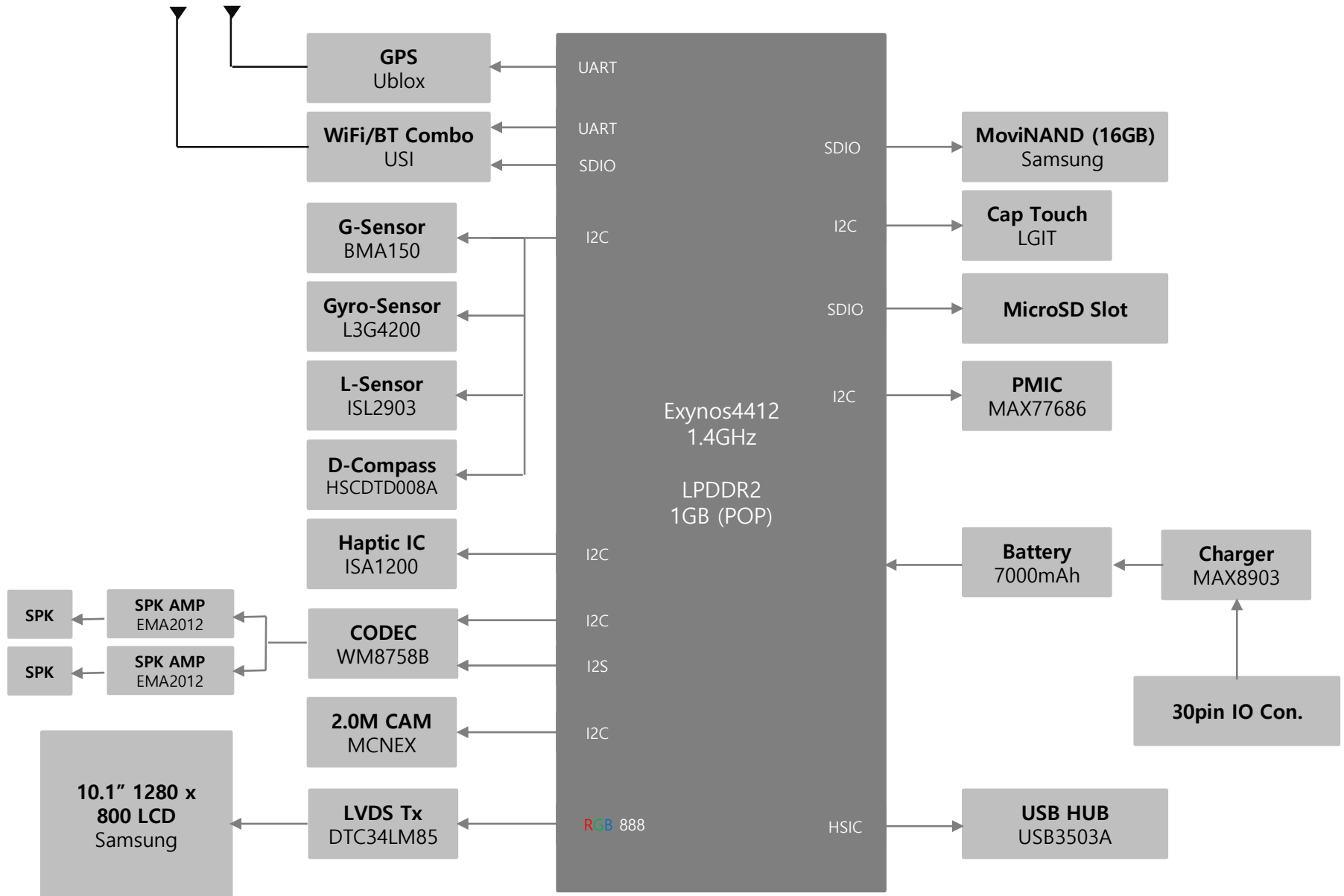
Single RF & Baseband IC, A-GPS

Main components

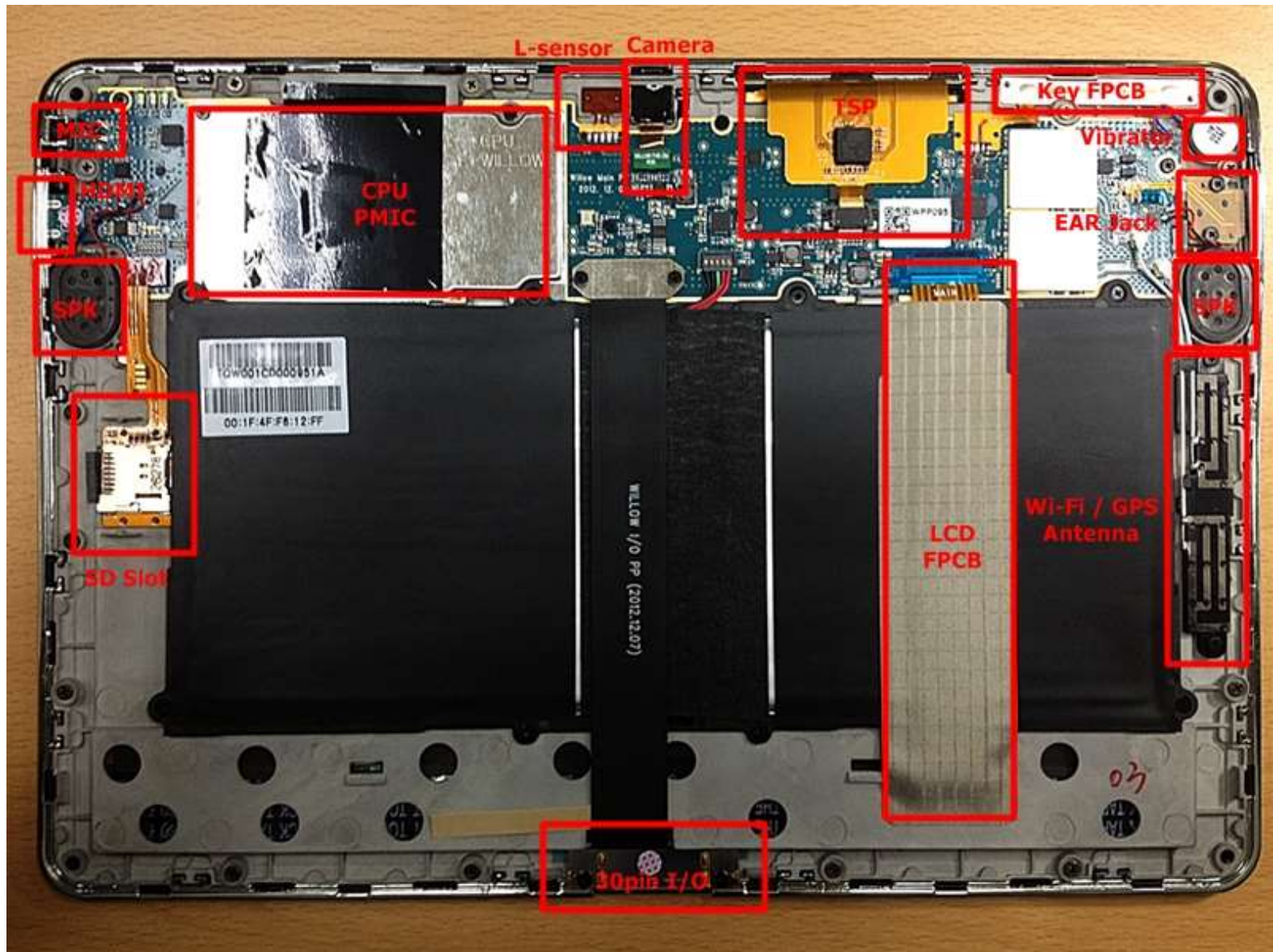
Item	Part #	Supplier
PMIC	MAX77686	MAXIM
Charger	MAX8903CETI+	MAXIM
Codec	WM8758BGEFL/V	Wolfson
Fuel Gauge	MAX17040G+	MAXIM
Haptic	ISA1200	Imagis
G-Sensor	BMA150	BOSCH
Gyro-Sensor	L3G4200	ST Micro
D-Compass	HSCDTD008A	ALPS
L-Sensor	ISL9203IROZ-T7	Intersil
LCD	LTN101AL03-801	SAMSUNG
Camera	WILLOW_2M	MCNEX
Speaker	HSO-2014BL20	Hea-Sung Techbig (혜성테크빅)
Battery	PR-37849N	TCL
Adaptor	SDCII10A	Shun Shing

HW Compositions

Block Diagram

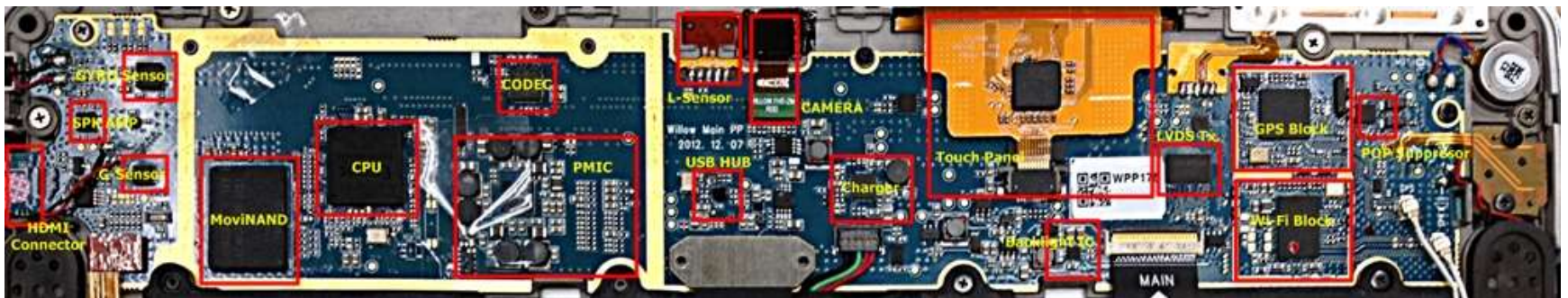


Composition



How to repair

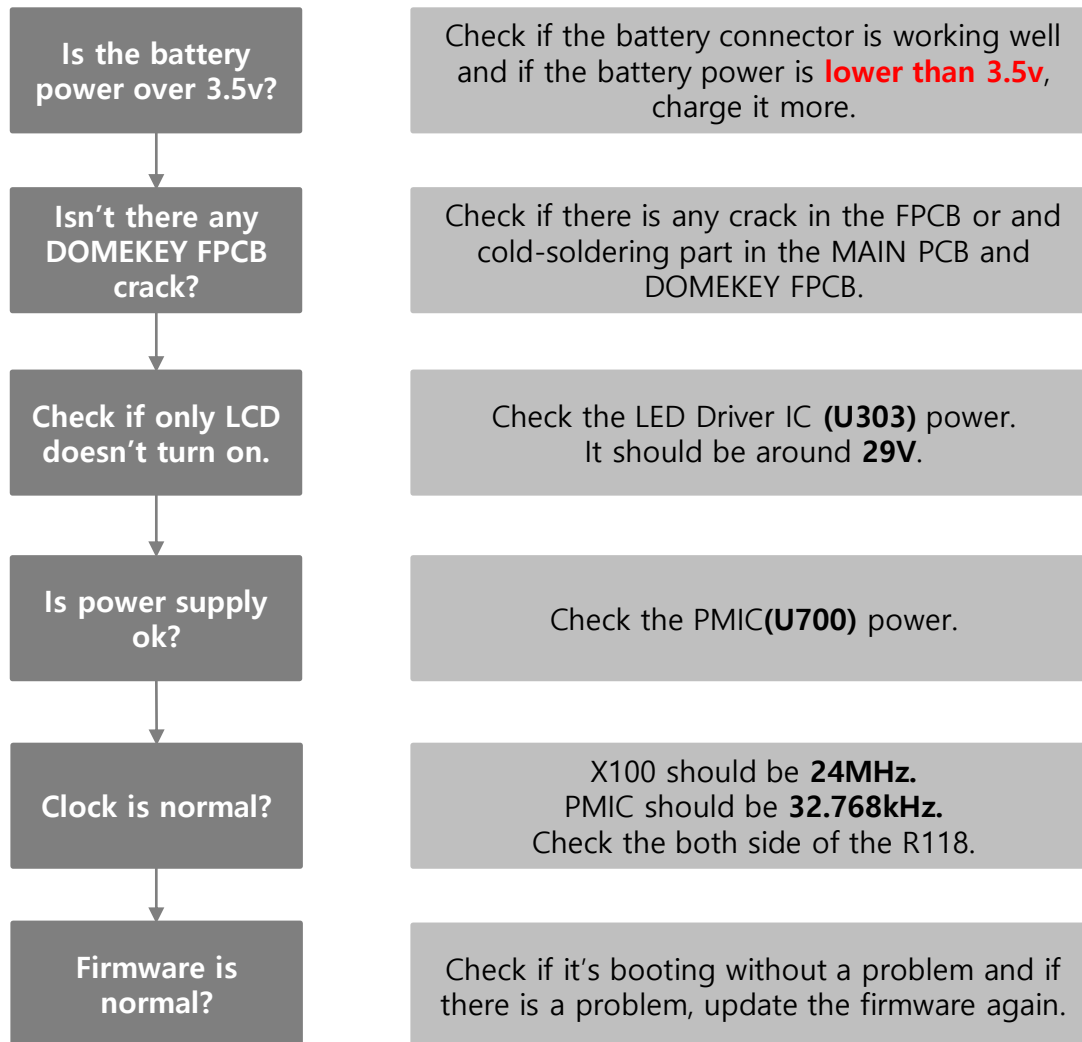
Main Block of main board



Main faulty

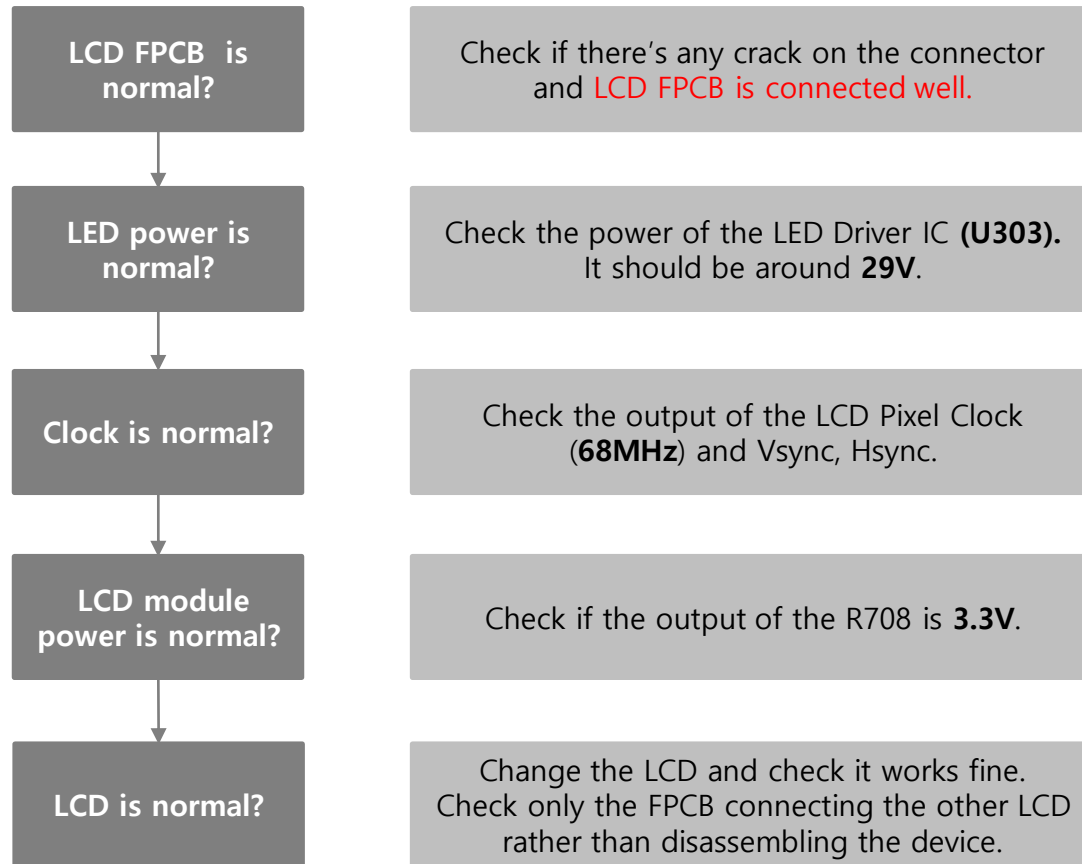
1. In case Power doesn't turn on
2. In case LCD doesn't turn on
3. In case a device is not charging
4. In case a Touch Screen doesn't work
5. In case a Speaker doesn't work
6. In case an Earphone doesn't work
7. In case a MiC doesn't work
8. In case a Micro SD doesn't work
9. In case a Camera doesn't work
10. In case Wi-Fi isn't connected
11. In case GPS doesn't work
12. In case sensors don't work

1. In case Power doesn't turn on



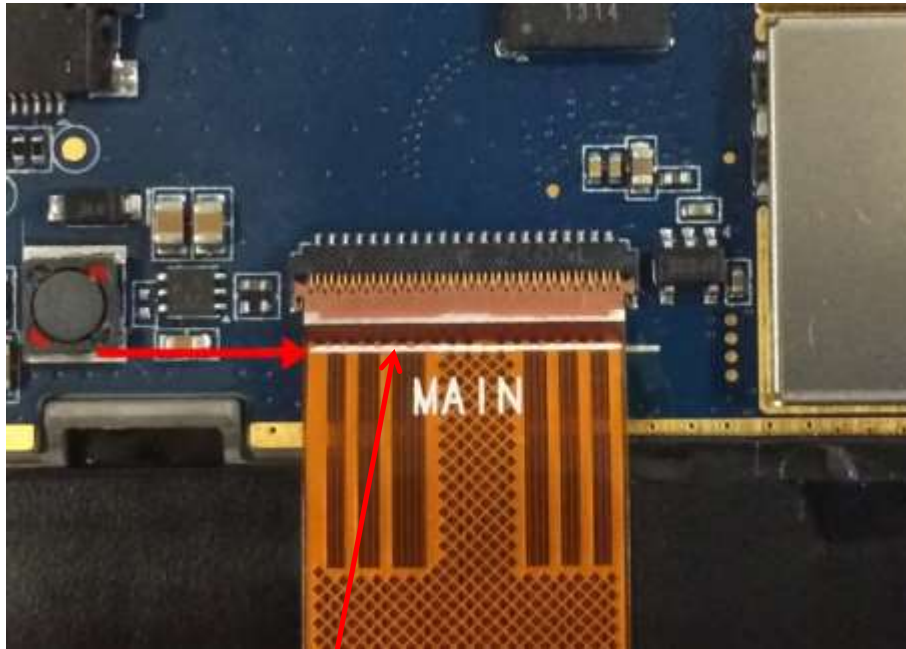
**Set only power on with adaptor. Set is not power on without adaptor.
→ This problem is that Battery Connector is not properly connected.**

2. In case LCD doesn't turn on

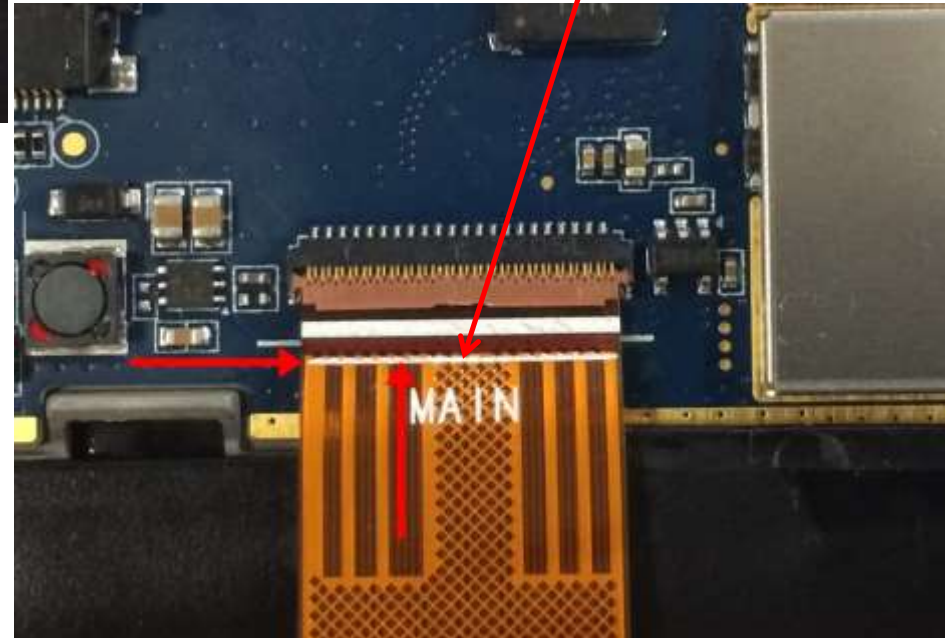


LCD FPCB is
normal?

Check if there's any crack on the connector
and LCD FPCB is connected well.



This is normal.



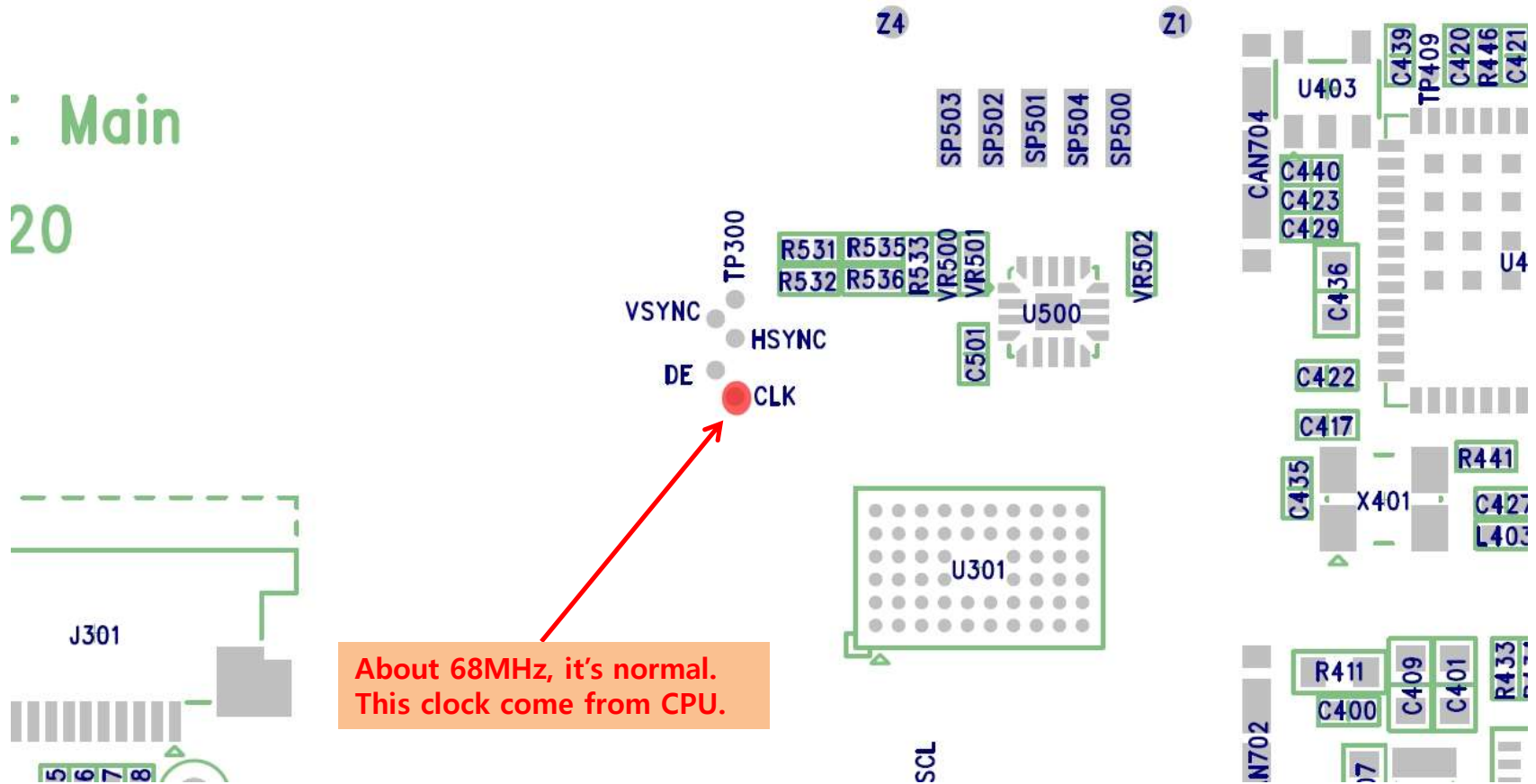
This is fail.

Clock is normal?

Check the output of the LCD Pixel Clock (68MHz) and Vsync, Hsync.

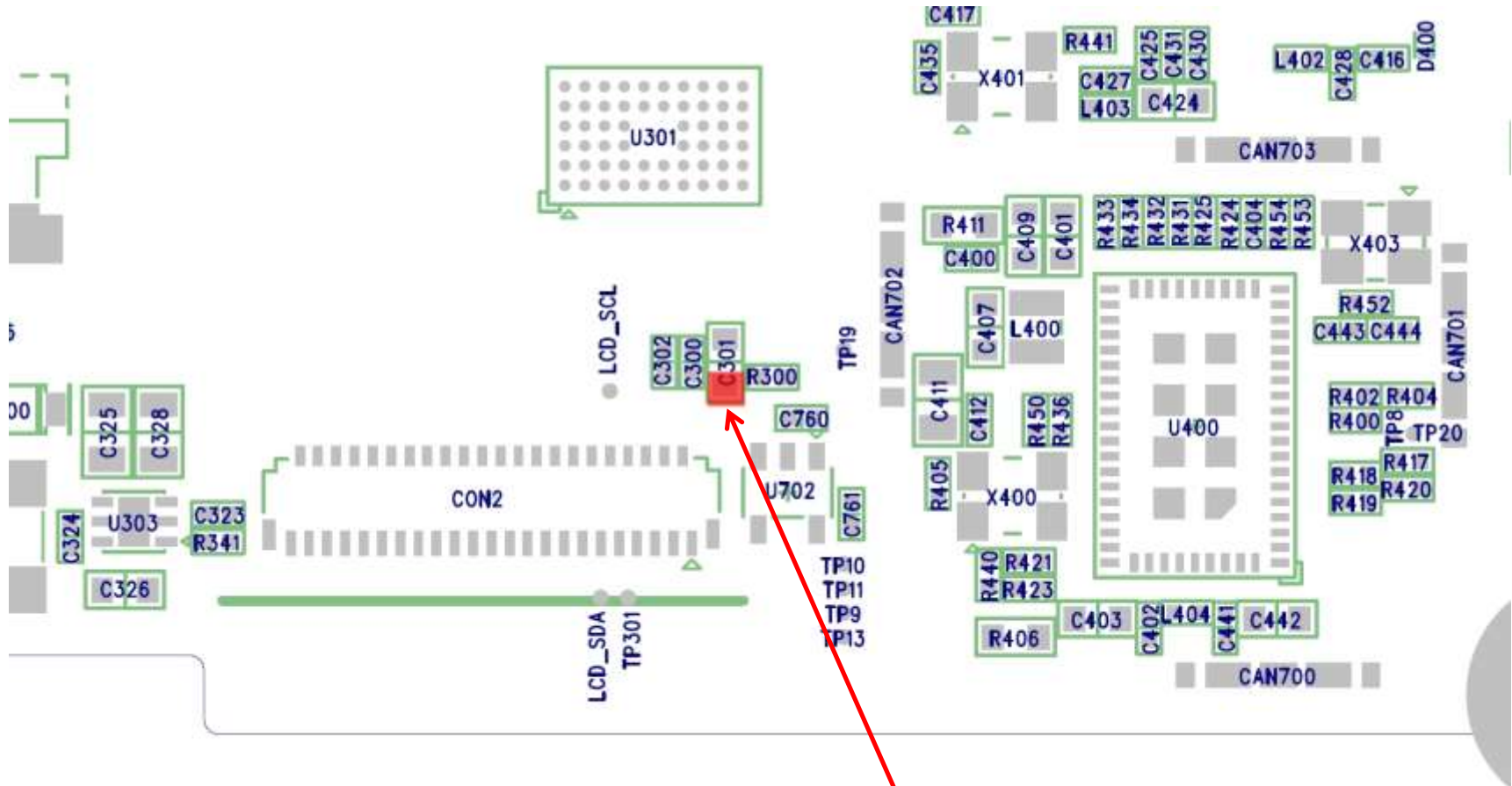
: Main

20



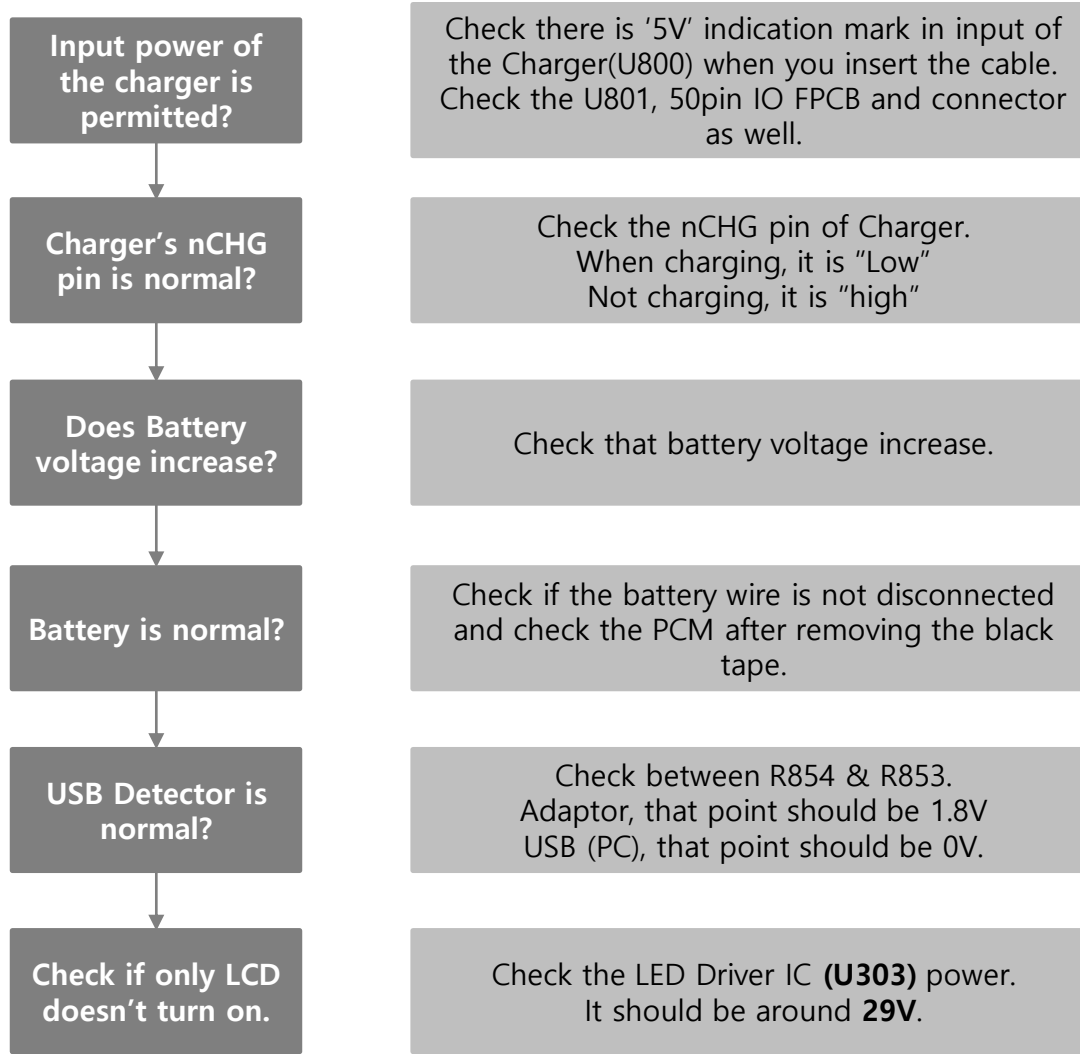
LCD module
power is normal?

Check if the output of the R708 is **3.3V**.



Around 3.3V, LCD power is OK.
This point is equal to R708 point.

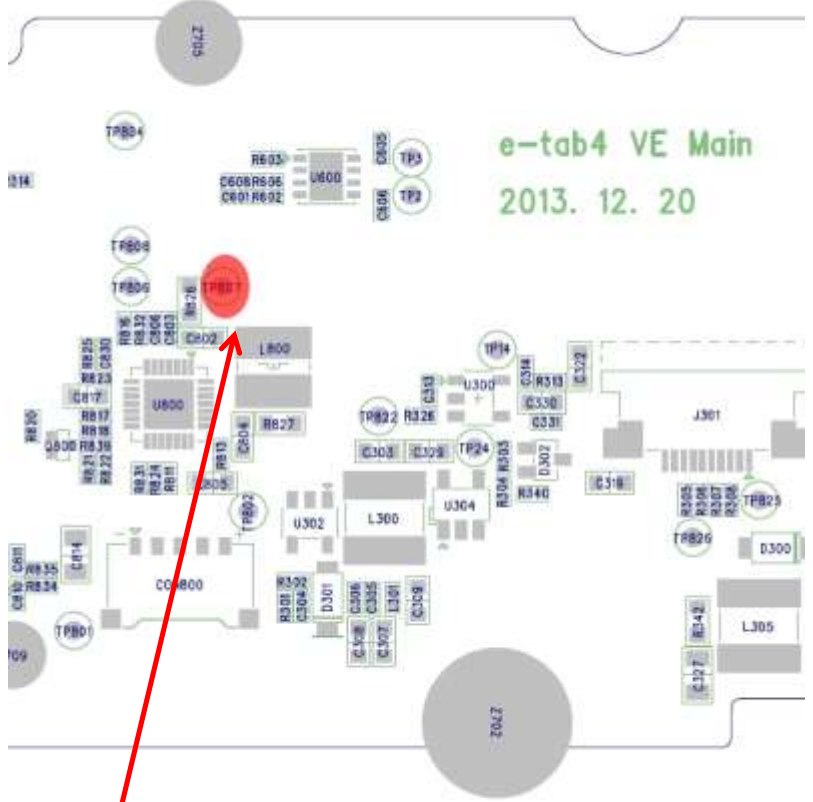
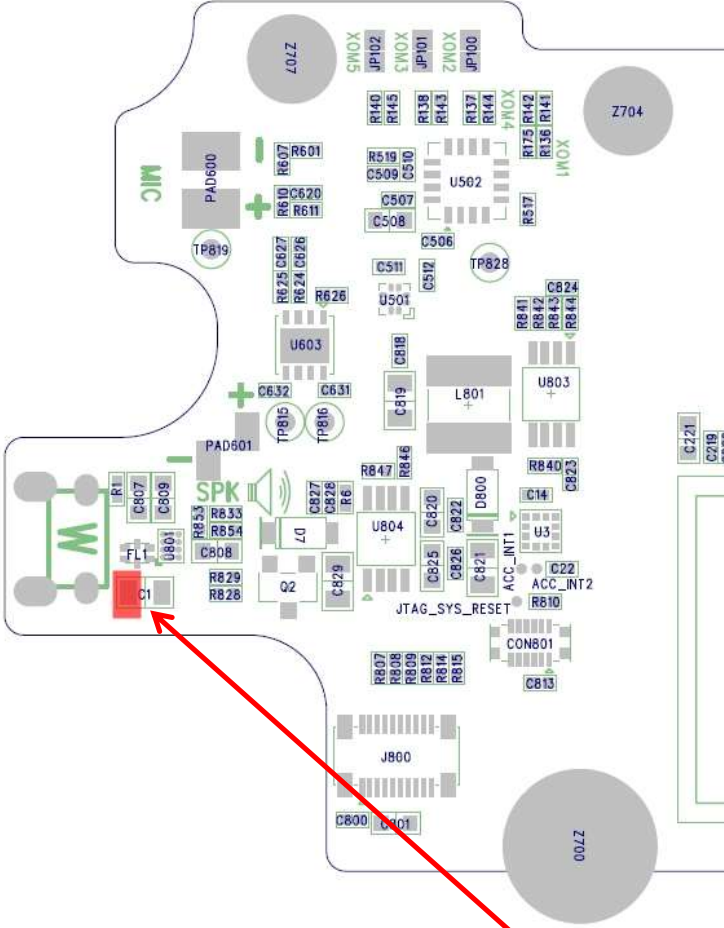
3. In case a device is not charging



Check 6,7,8pin of U804. If there is strange voltage at that point, U804 is defective. It would be cause Not Charging.

Input power of the charger is permitted?

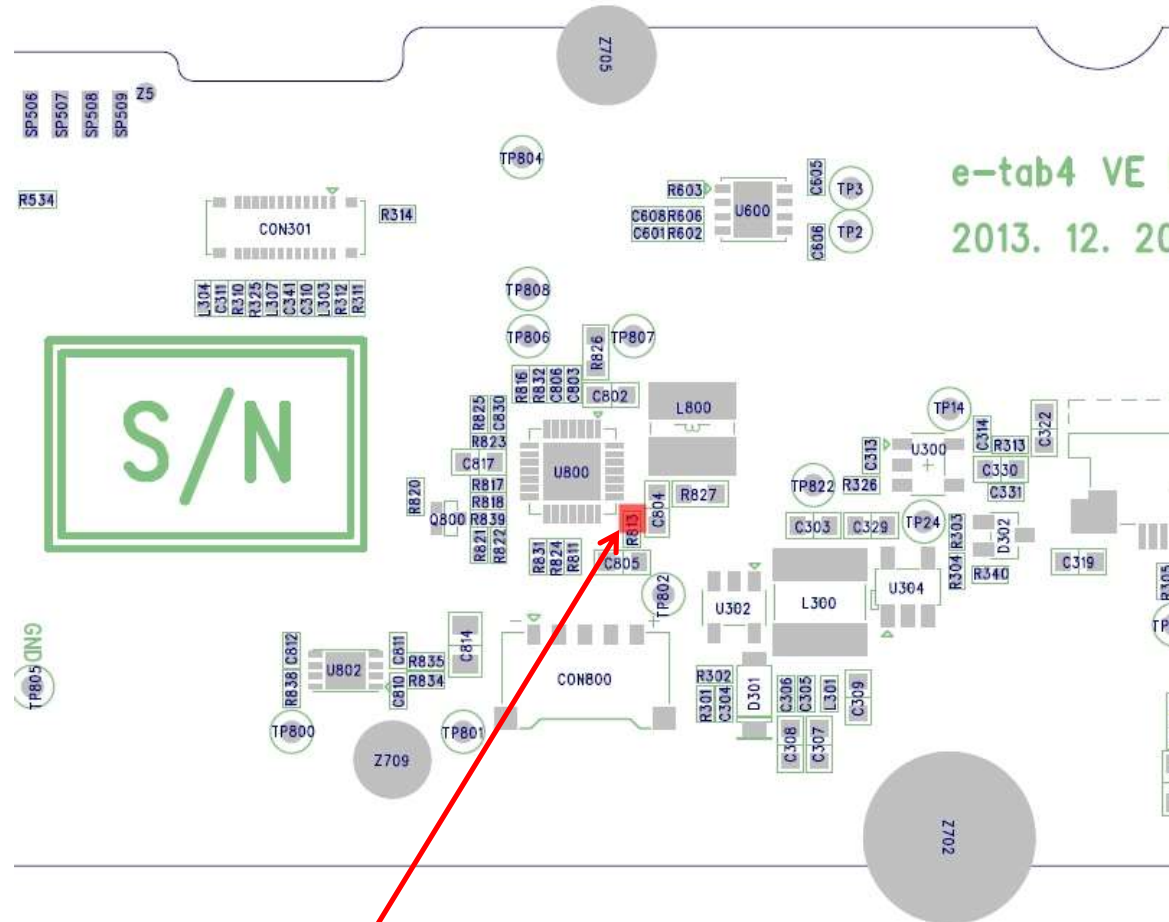
Check there is '5V' indication mark in input of the Charger(U800) when you insert the cable.
Check the U801, 50pin IO FPCB and connector as well.



This point should be around 5V.

Charger's nCHG pin is normal?

Check the nCHG pin of Charger.
When charging, it is "Low"
Not charging, it is "high"



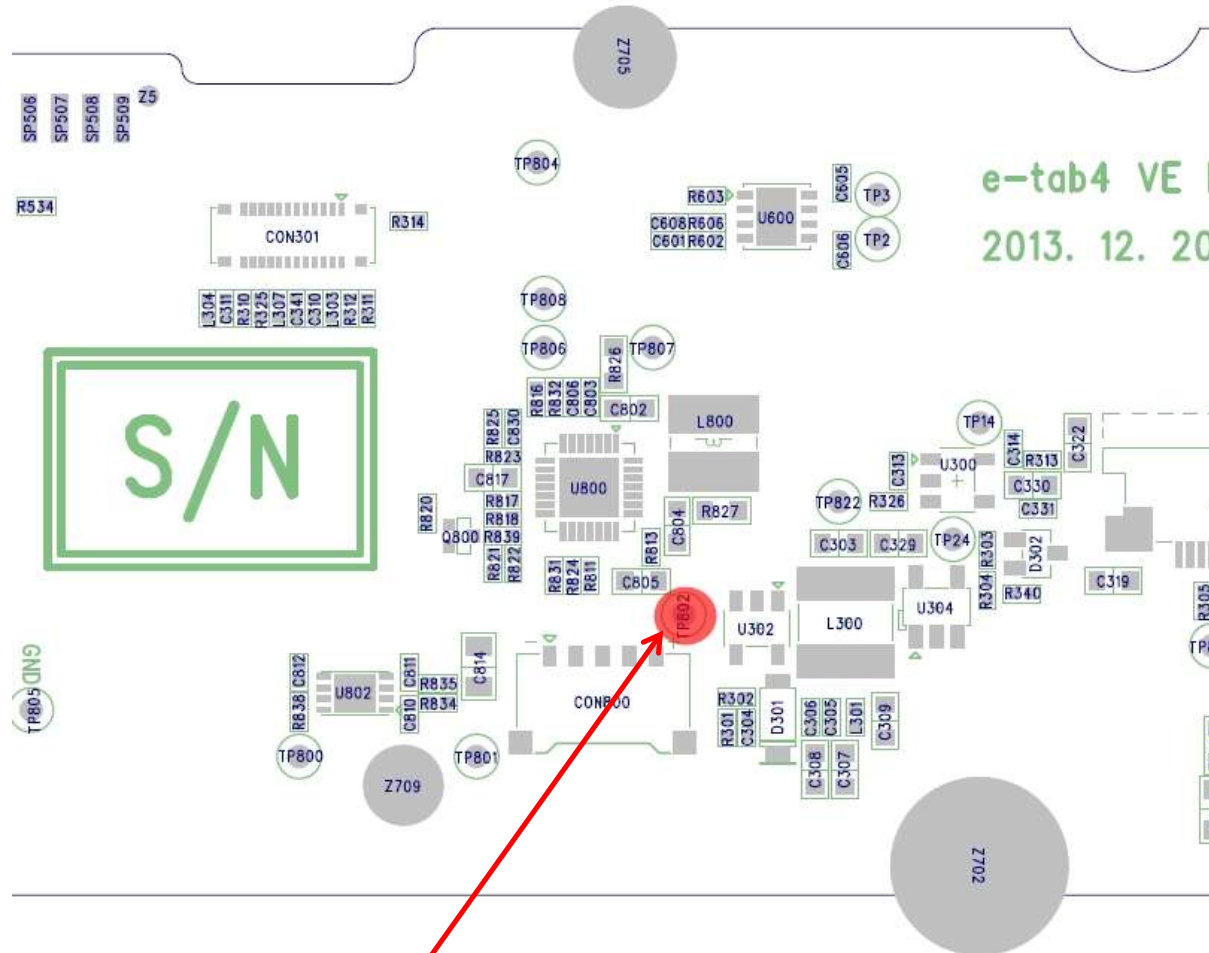
e-tab4 VE I
2013. 12. 20

S/N

This point should be around 0V.

Does Battery
voltage increase?

Check that battery voltage increase.



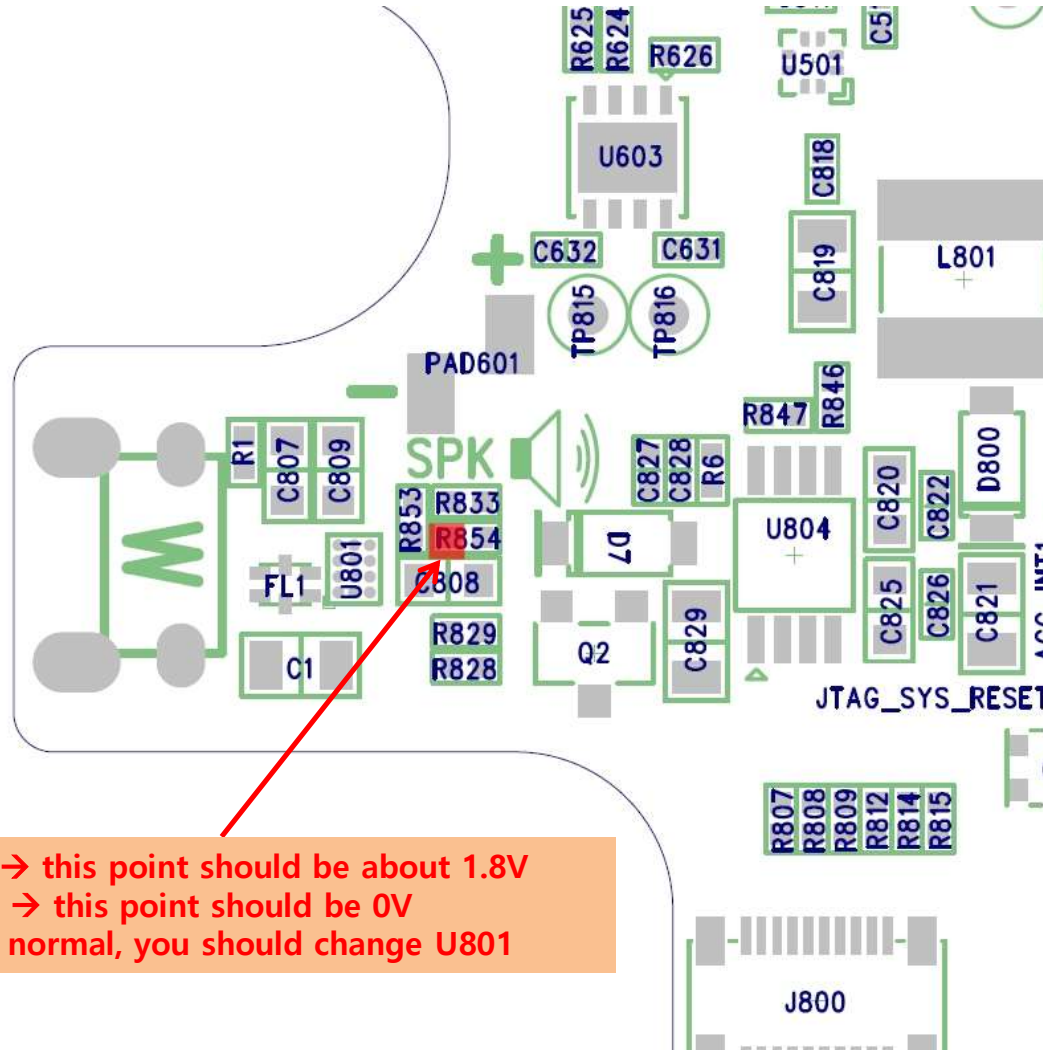
e-tab4 VE I
2013. 12. 20

S/N

This point voltage should increase.

USB Detector is
normal?

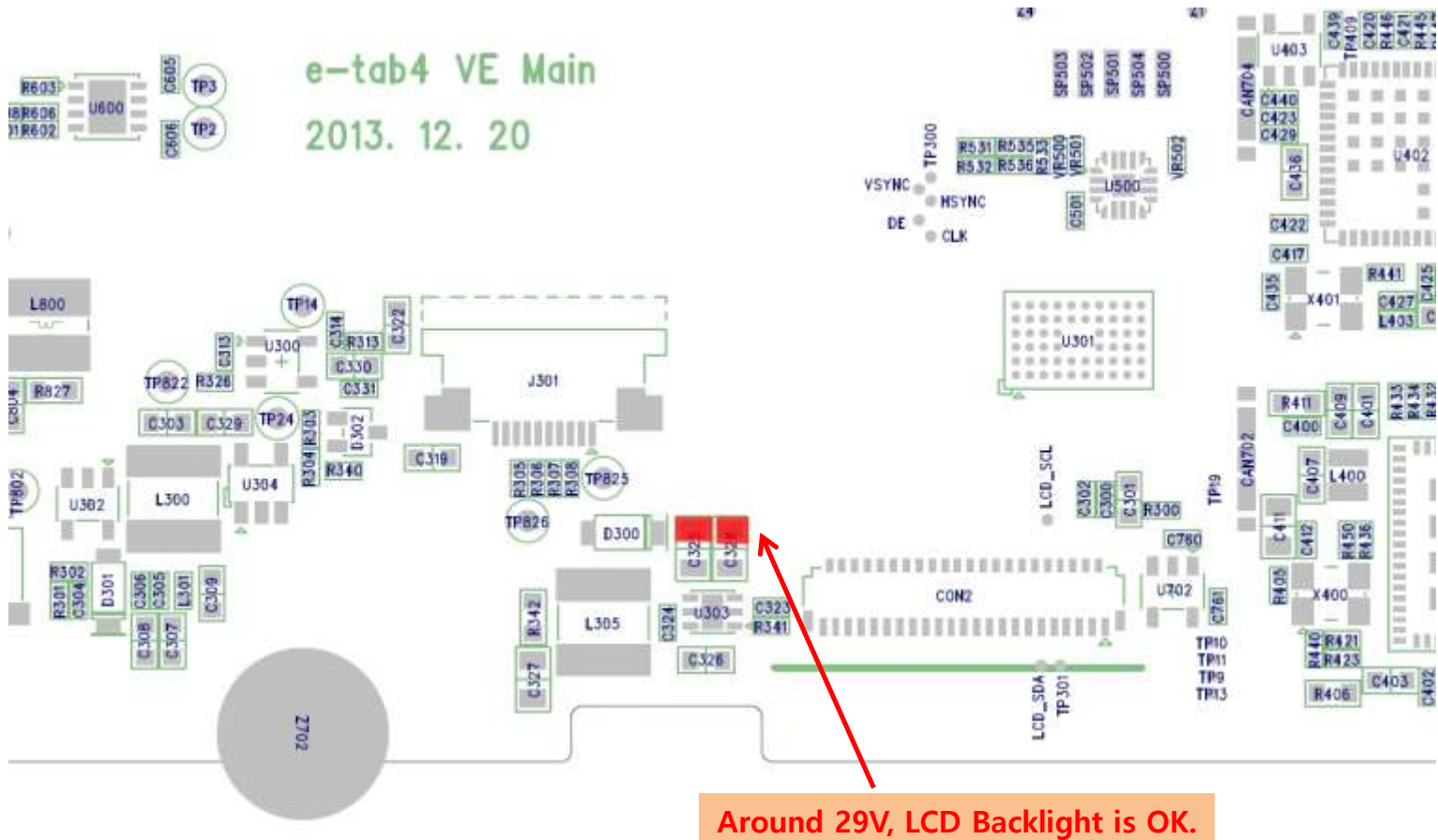
Check between R854 & R853.
Adaptor, that point should be 1.8V
USB (PC), that point should be 0V.



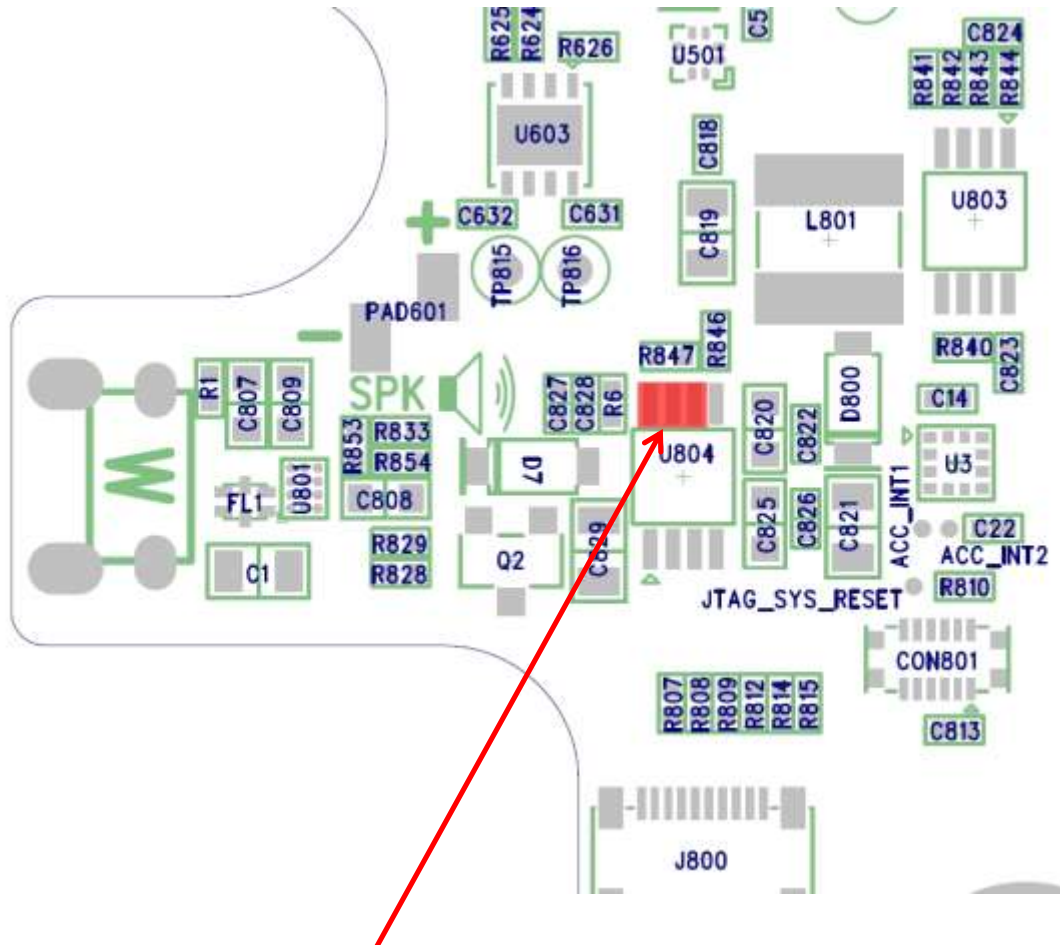
Adaptor charging → this point should be about 1.8V
USB (PC) charging → this point should be 0V
If this point is not normal, you should change U801

Check if only LCD
doesn't turn on.

Check the LED Driver IC (**U303**) power.
It should be around **29V**.

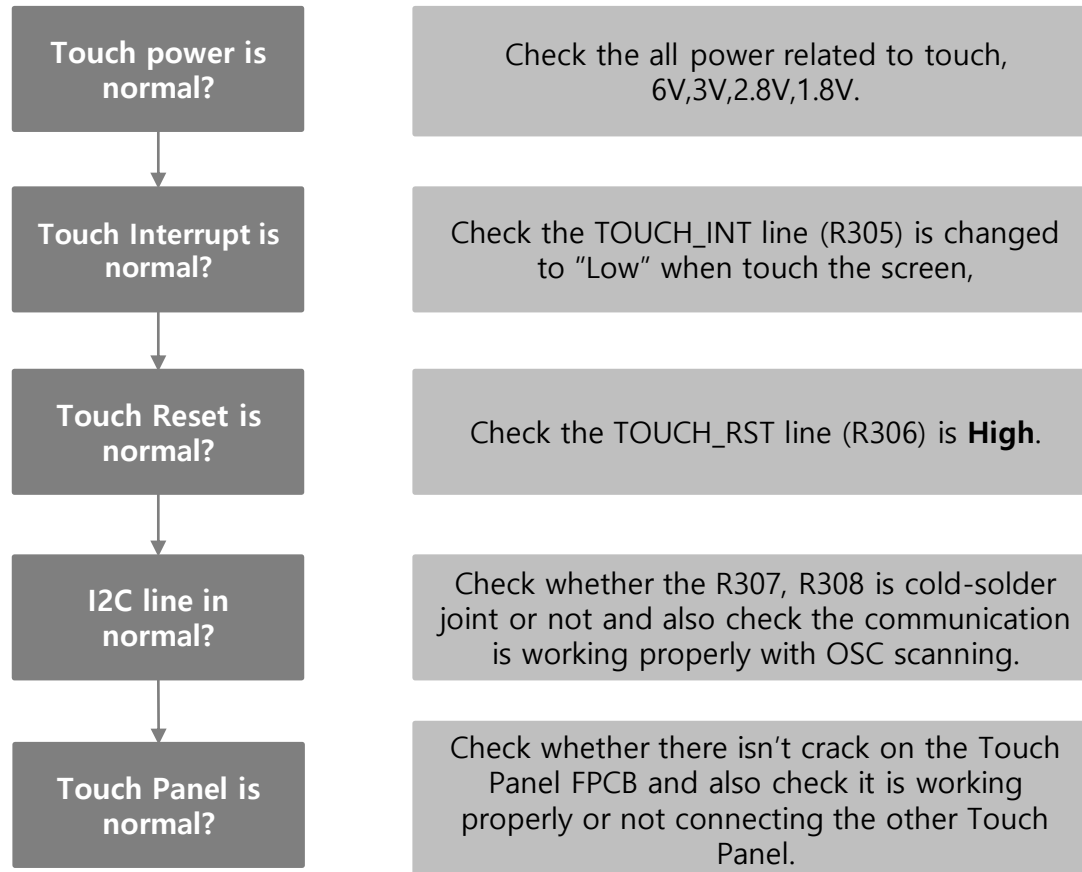


Check 6,7,8pin of U804. If there is strange voltage at that point, U804 is defective. It would be cause Not Charging.



This point must be 0V.
This point is 5V only USB HOST Mode.
When charging, if this point is strange voltage,
U804 is defective.
Change U804.

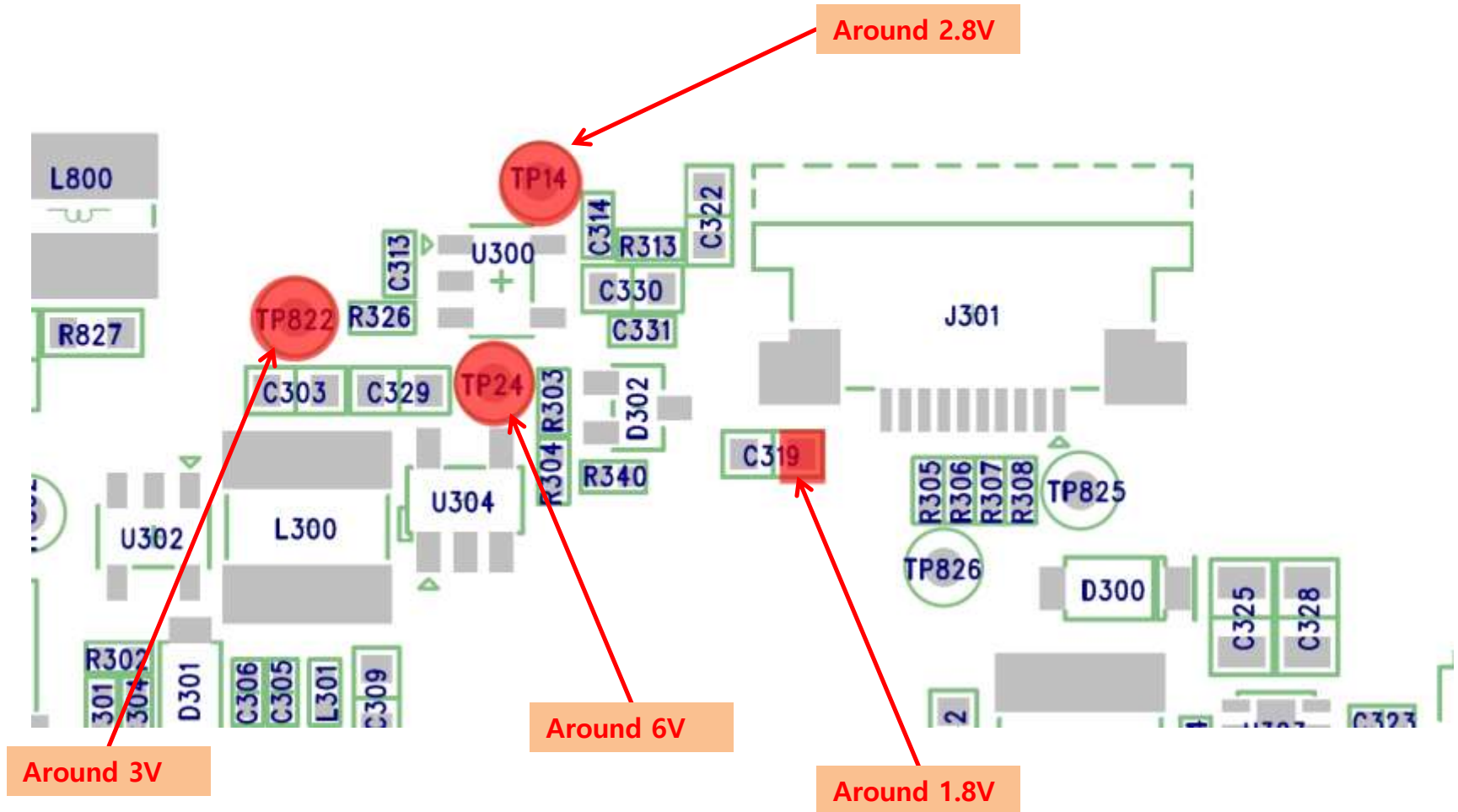
4. In case a Touch Screen doesn't work



**If only you touch two times, touch panel detect.
In this case, you should upgrade to MPTEST firmware.
And one more check.**

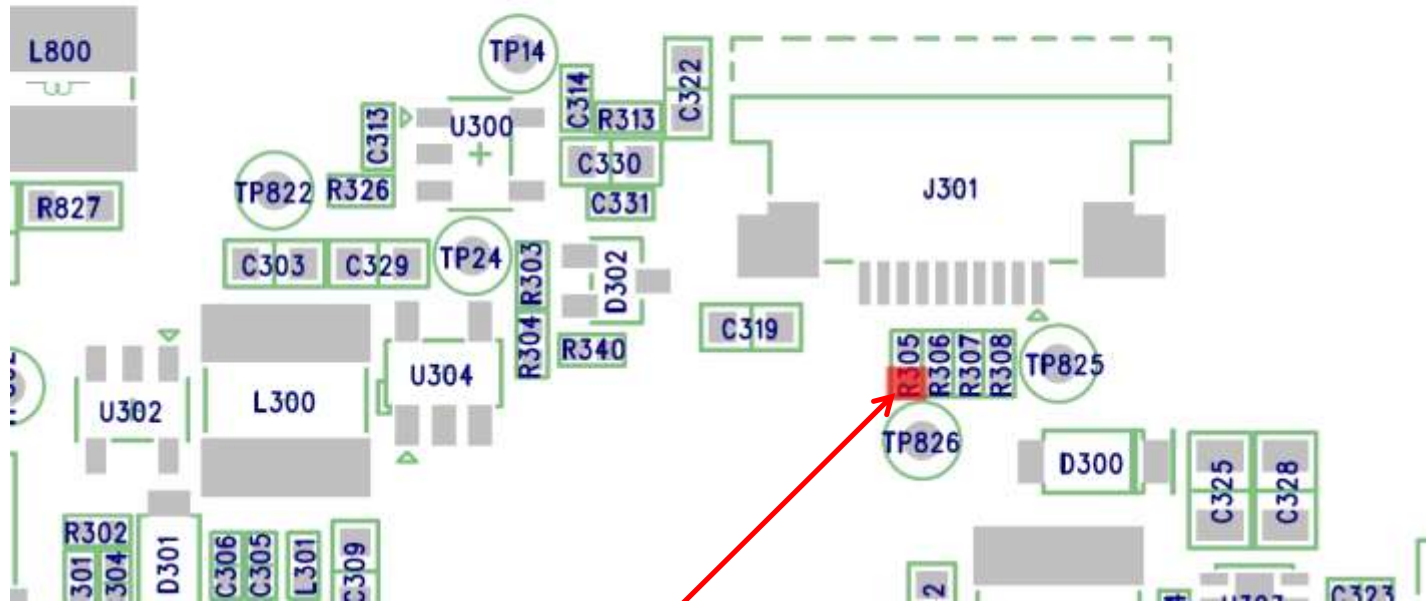
Touch power is normal?

Check the all power related to touch, 6V,3V,2.8V,1.8V.



Touch Interrupt is normal?

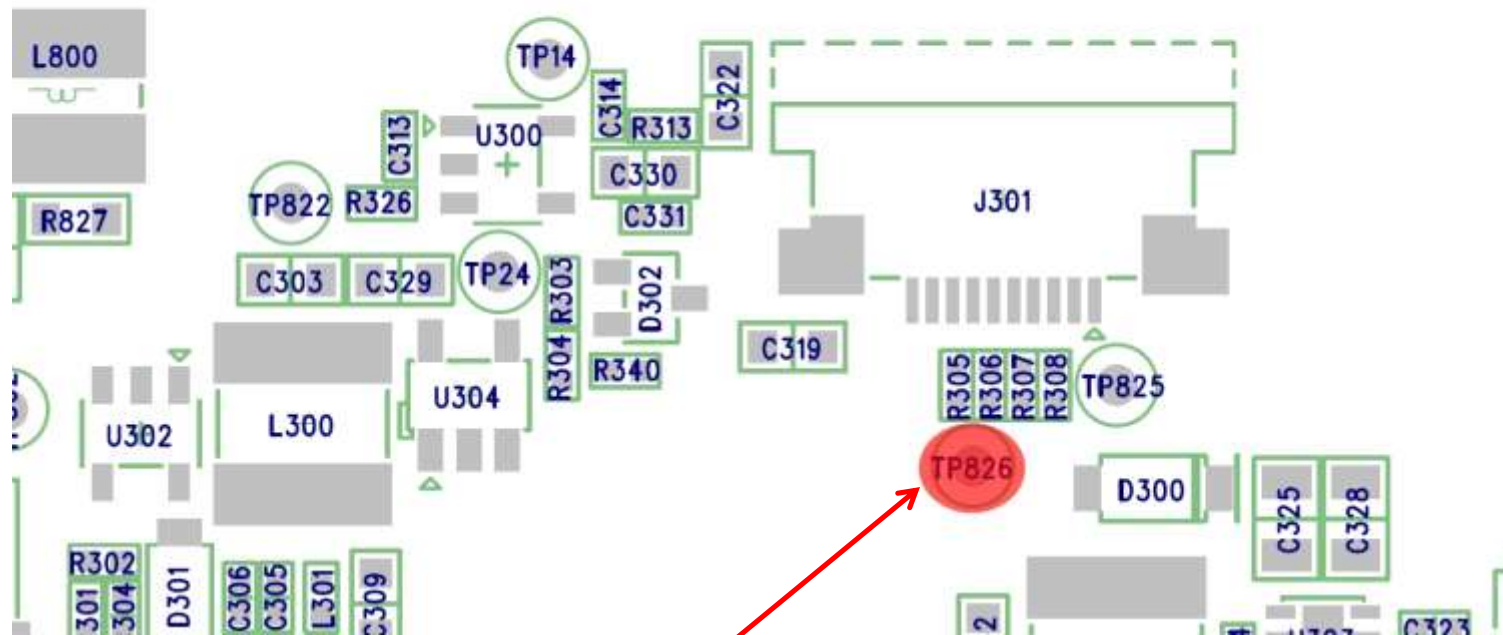
Check the TOUCH_INT line (R305) is changed to "Low" when touch the screen,



When not touch, this point keep 1.8V.
When touch, this point is changed Low.

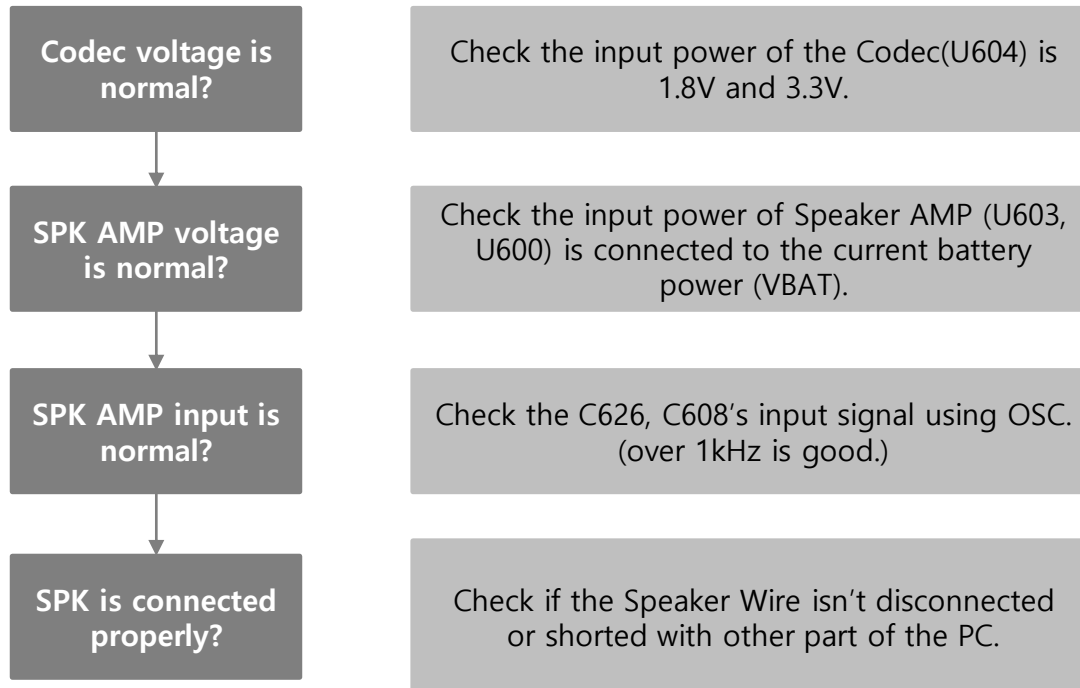
I2C line in normal?

Check whether the R307, R308 is cold-solder joint or not and also check the communication is working properly with OSC scanning.



This point is I2C data line.
When booting complete, this point should keep 1.8V.
If this point continuously change, you should upgrade MPTEST firmware.

5. In case a Speaker doesn't work

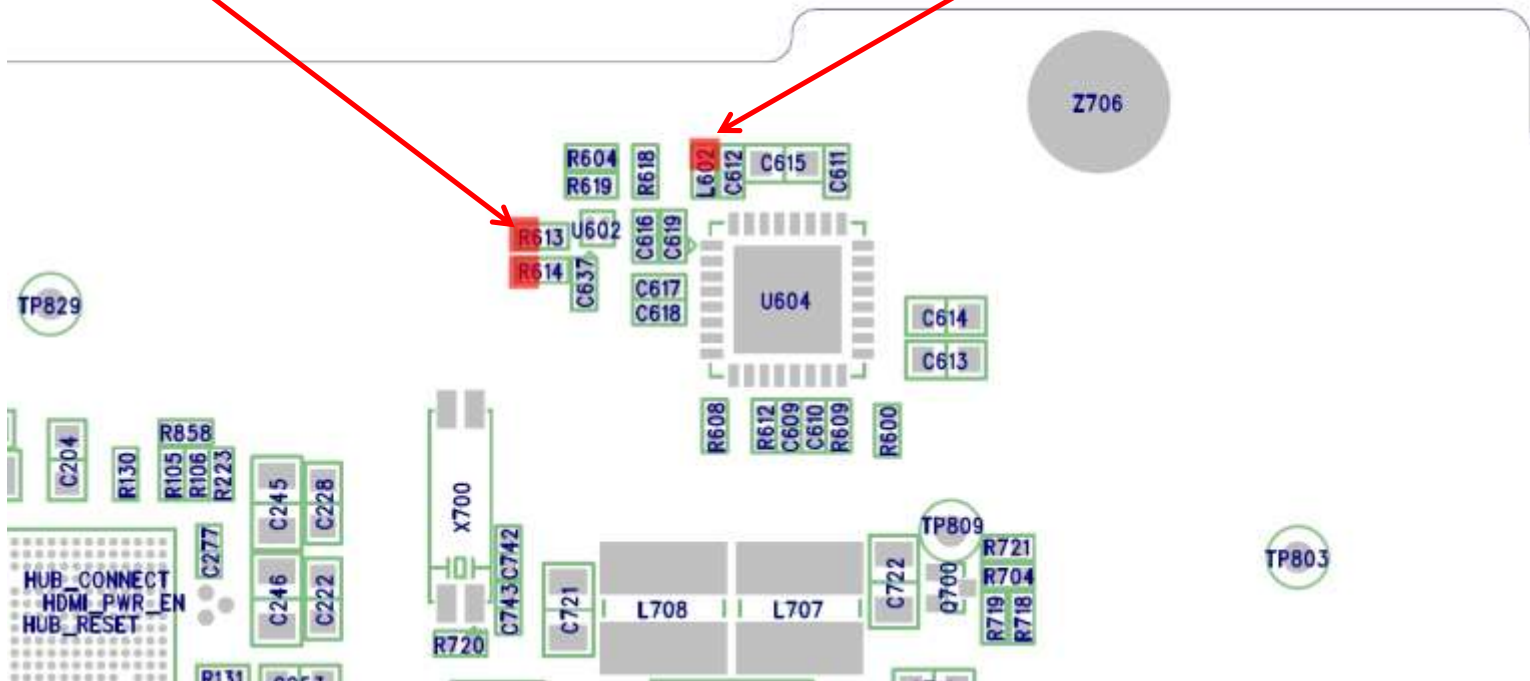


Codec voltage is normal?

Check the input power of the Codec(U604) is 1.8V and 3.3V.

Around 1.8V

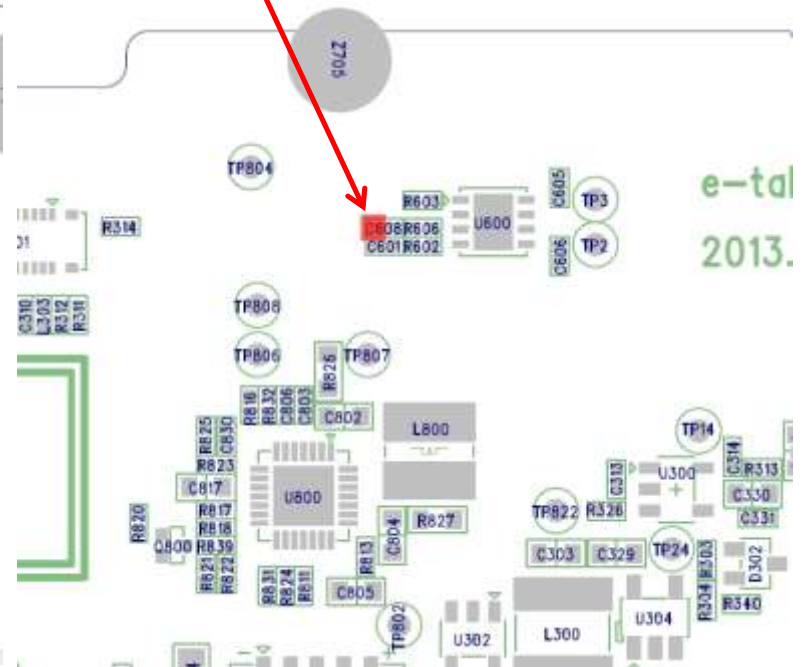
Around 3.3V



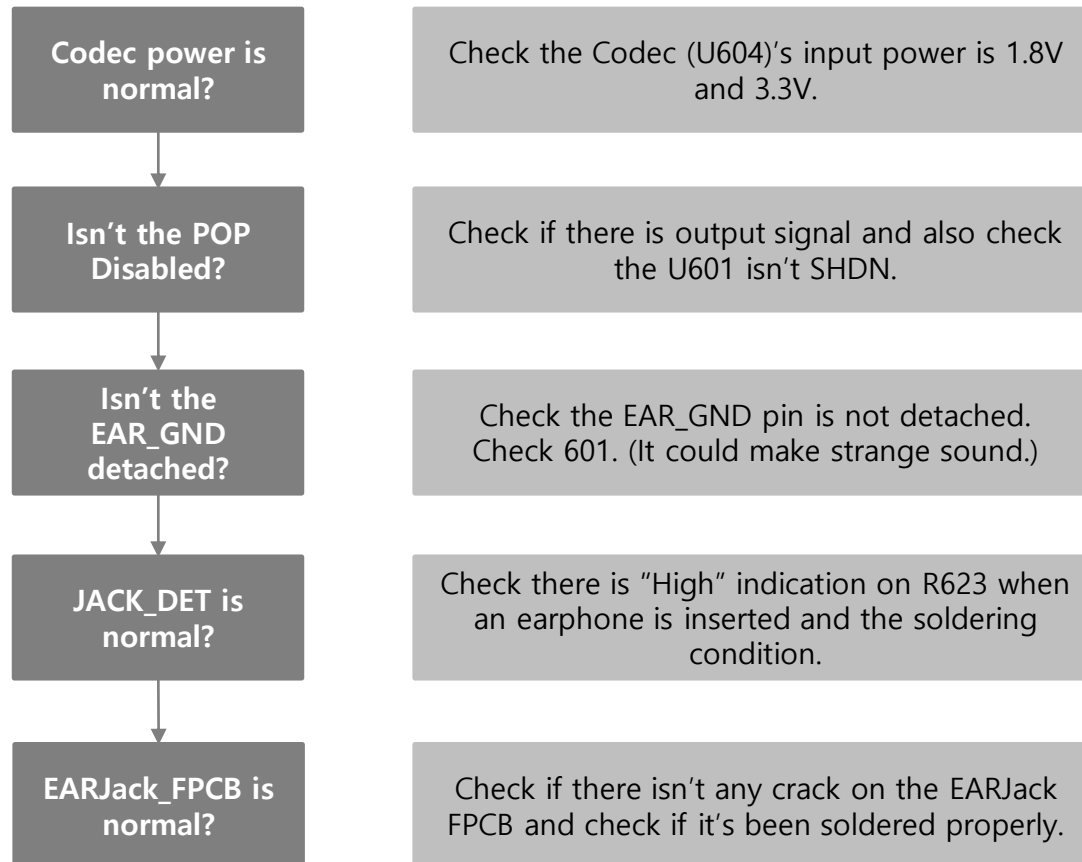
SPK AMP input is normal?

Check the C626, C608's input signal using OSC.
(over 1kHz is good.)

Check input signal of SPK Amp.



6. In case an Earphone doesn't work

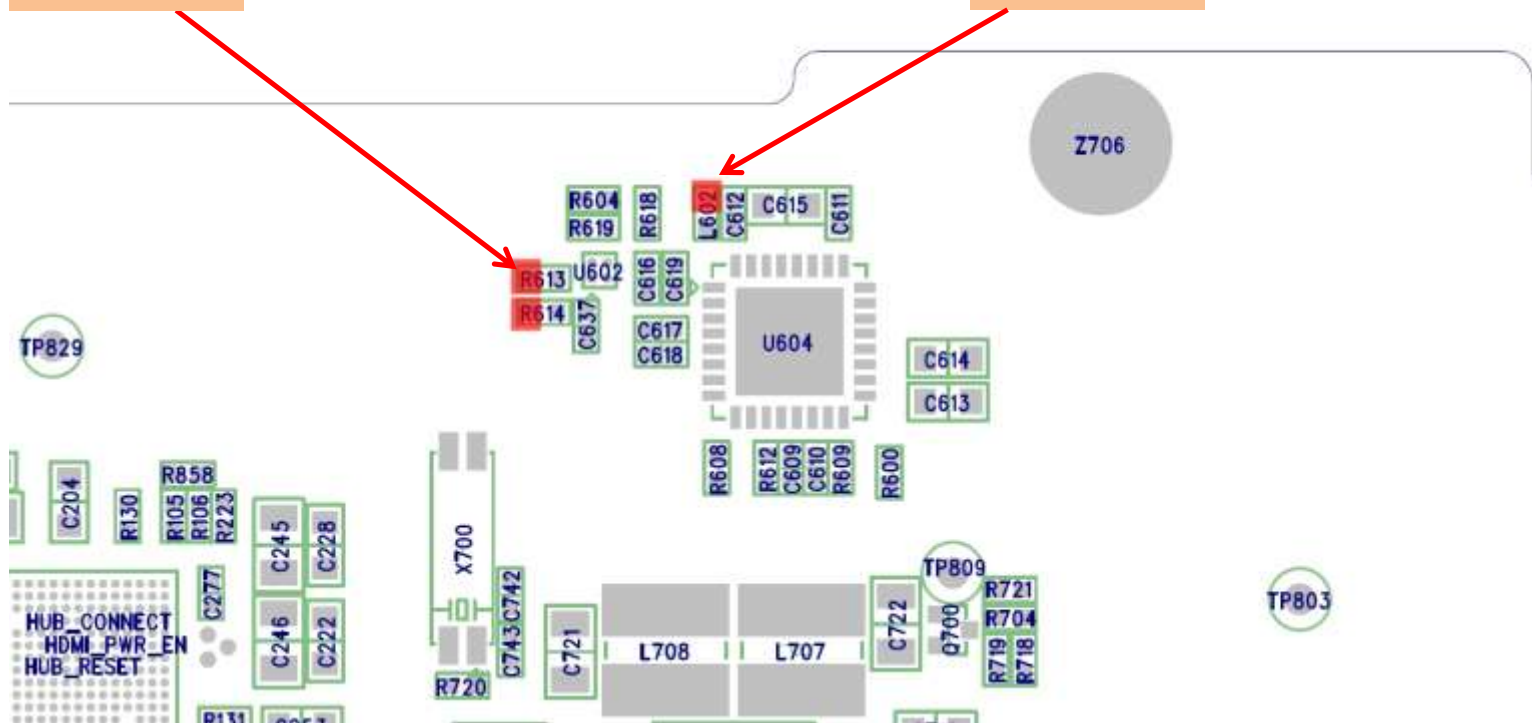


Codec voltage is normal?

Check the input power of the Codec(U604) is 1.8V and 3.3V.

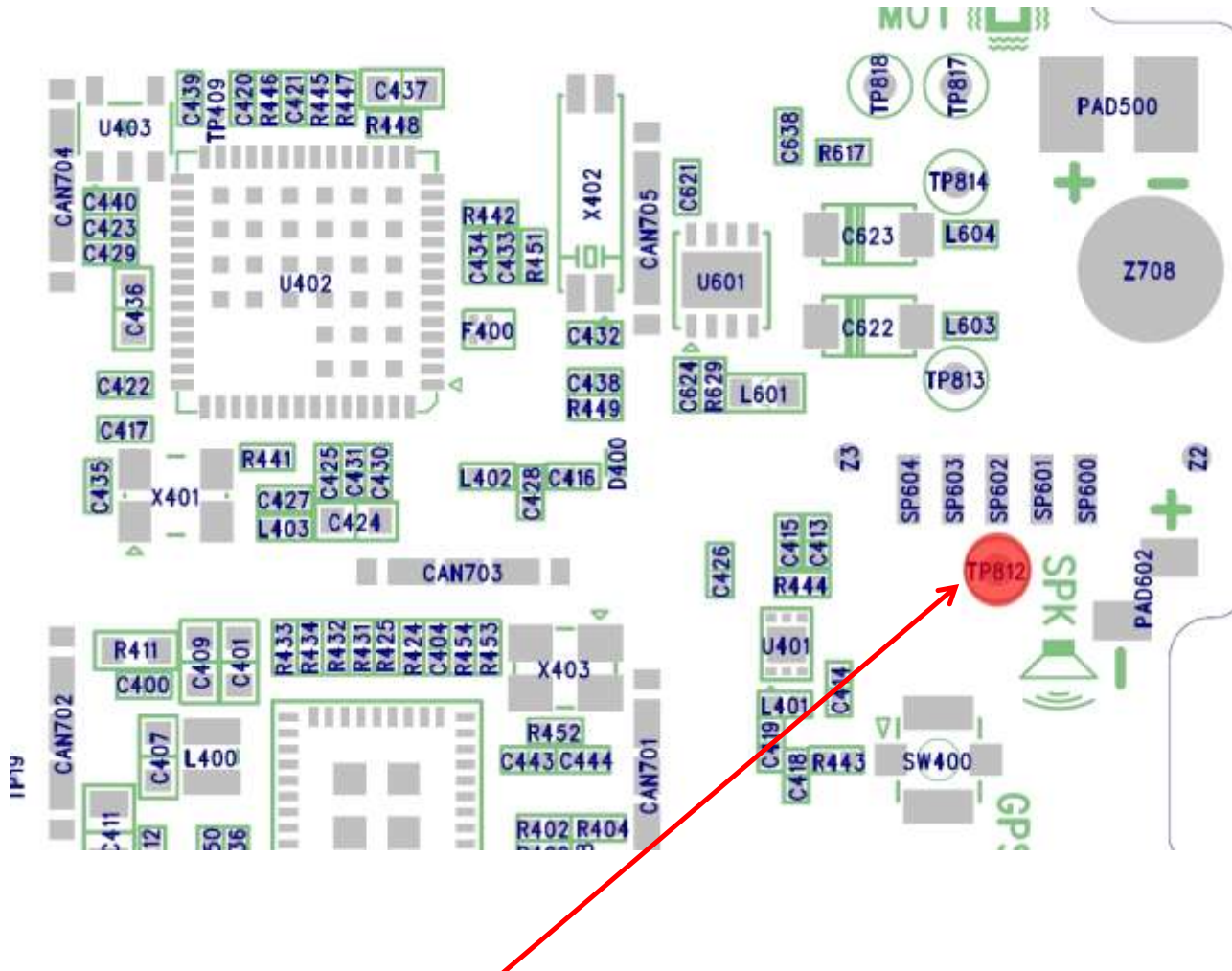
Around 1.8V

Around 3.3V



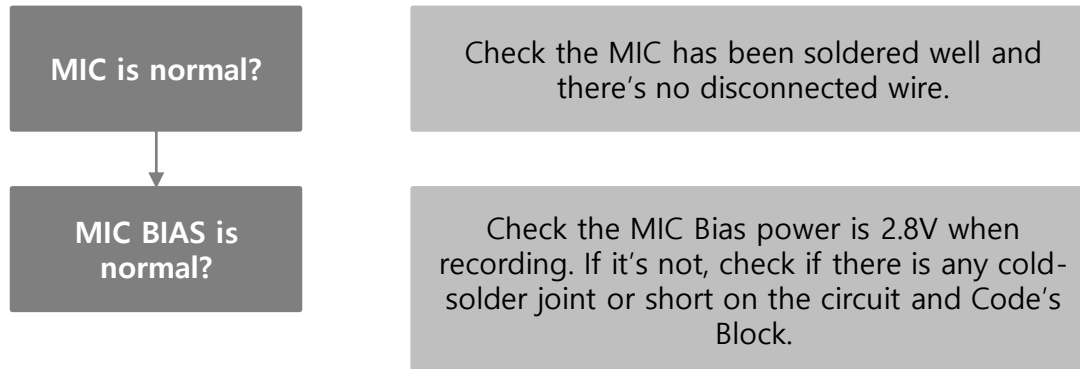
JACK_DET is
normal?

Check there is "High" indication on R623 when
an earphone is inserted and the soldering
condition.



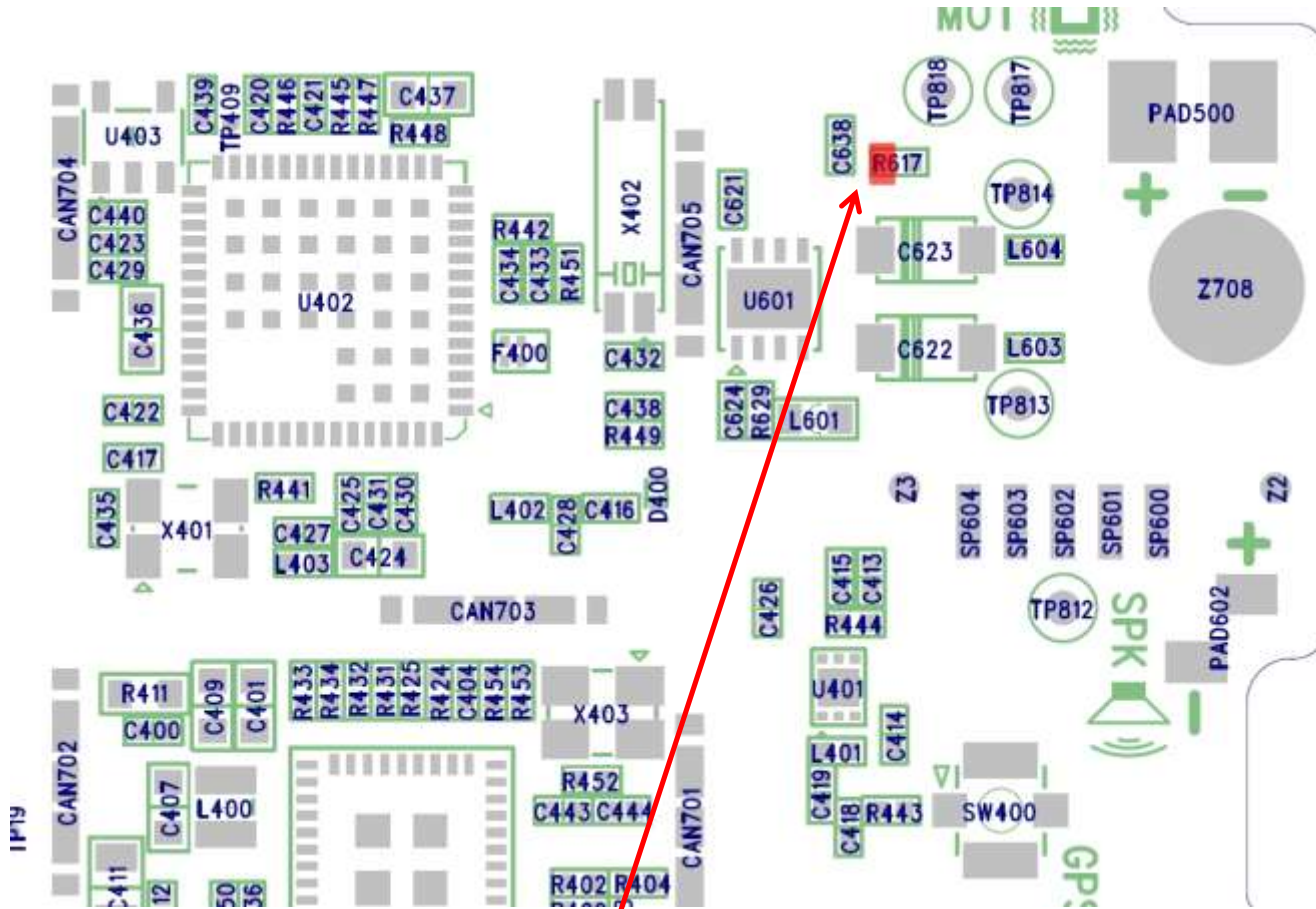
When earphone is connected, this point is about 1.8V.
If it is not normal, you should check Earjack FPCB.

7. In case a MiC doesn't work



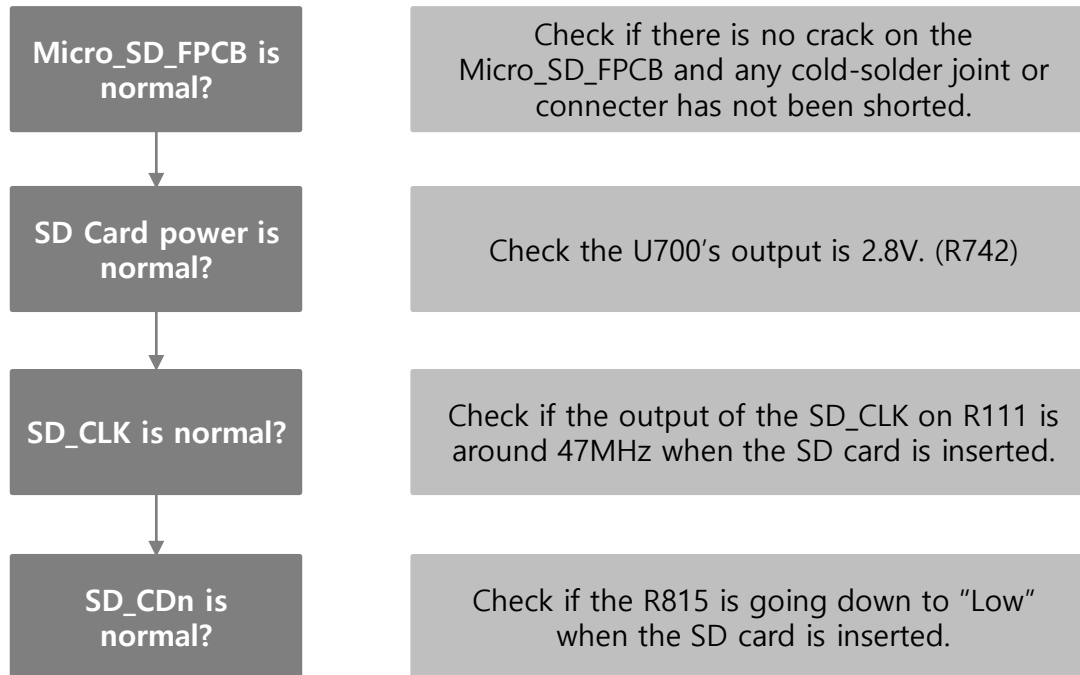
MIC BIAS is normal?

Check the MIC Bias power is 2.8V when recording. If it's not, check if there is any cold-solder joint or short on the circuit and Code's Block.



Around 2.8V
If this point is not normal, you should check Codec.

8. In case a Micro SD doesn't work



SD_CLK is normal?

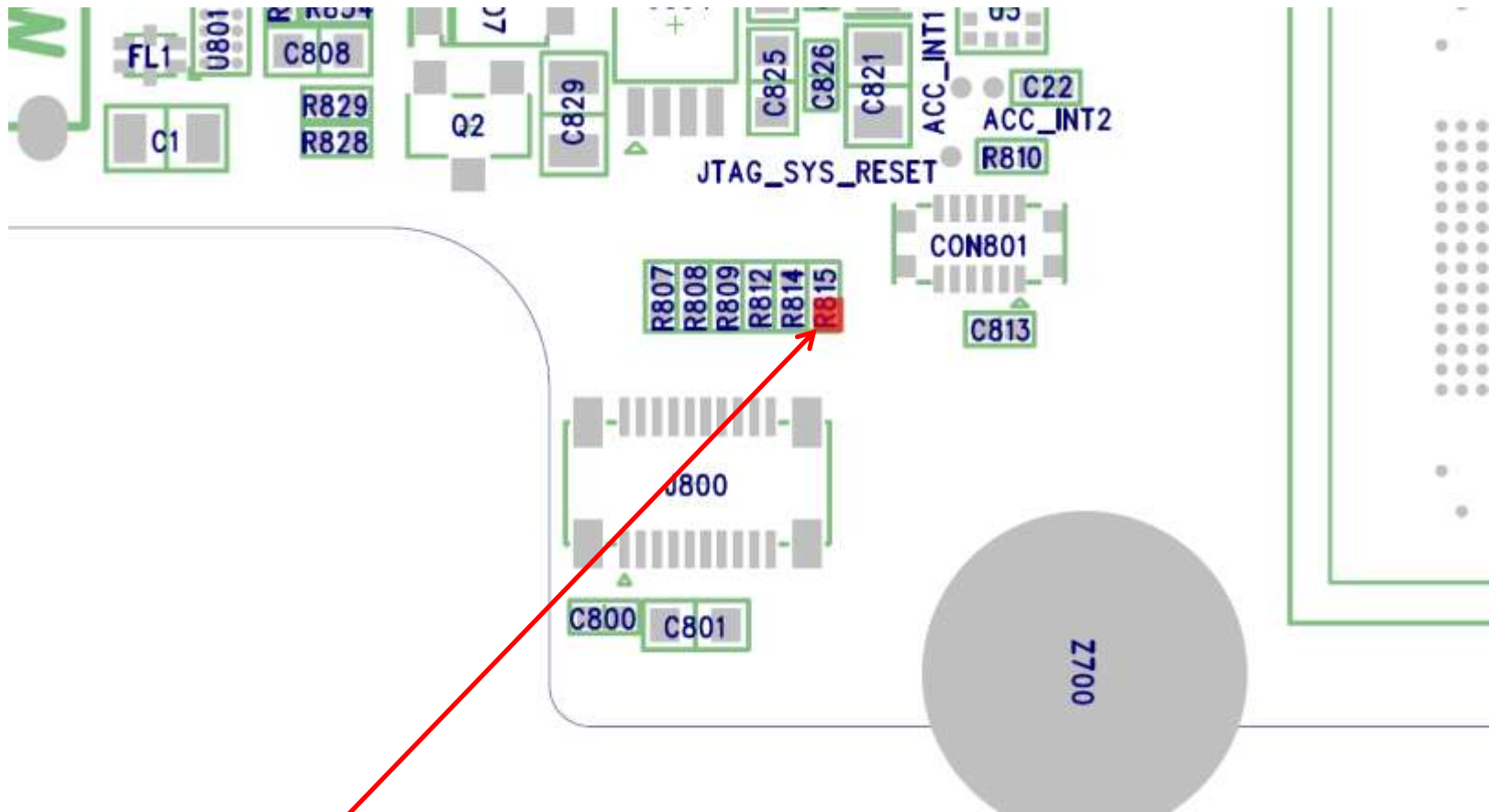
Check if the output of the SD_CLK on R111 is around 47MHz when the SD card is inserted.



About Clock is 47MHz

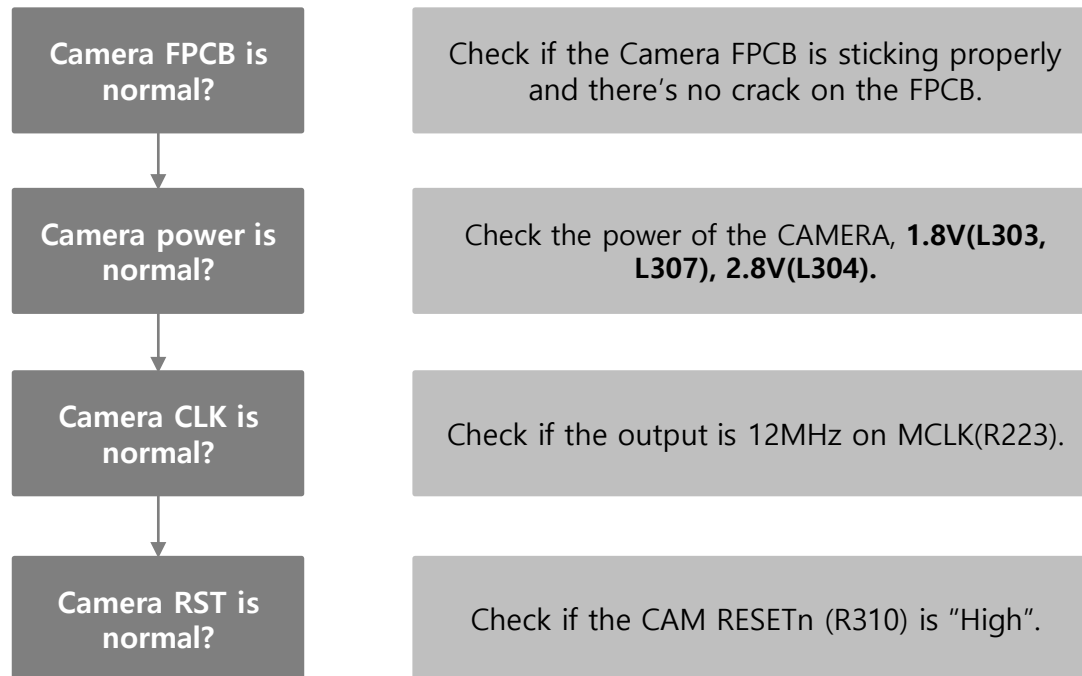
SD_CDn is
normal?

Check if the R815 is going down to "Low"
when the SD card is inserted.



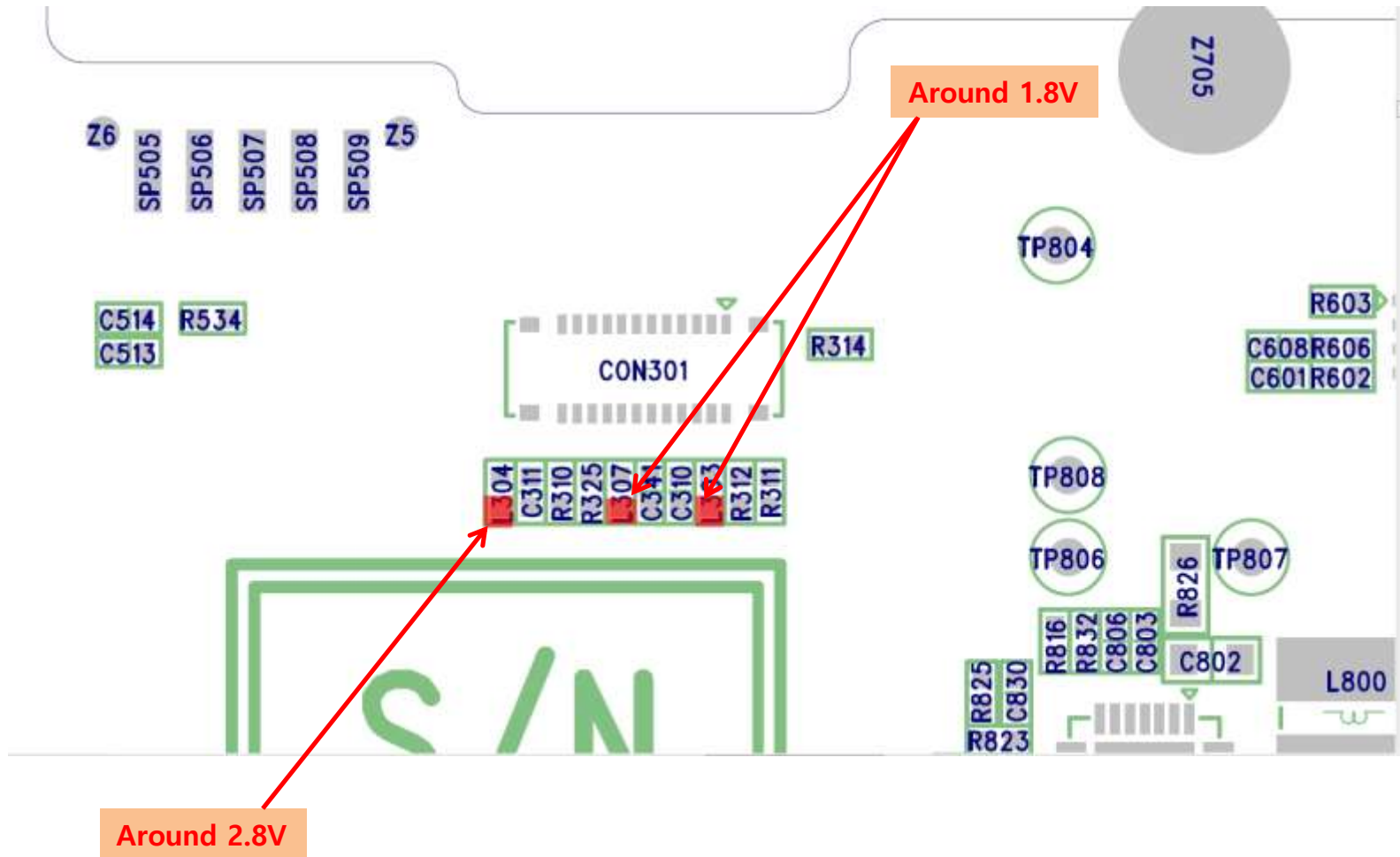
When the SD Card is inserted, this point is 0V

9. In case a Camera doesn't work



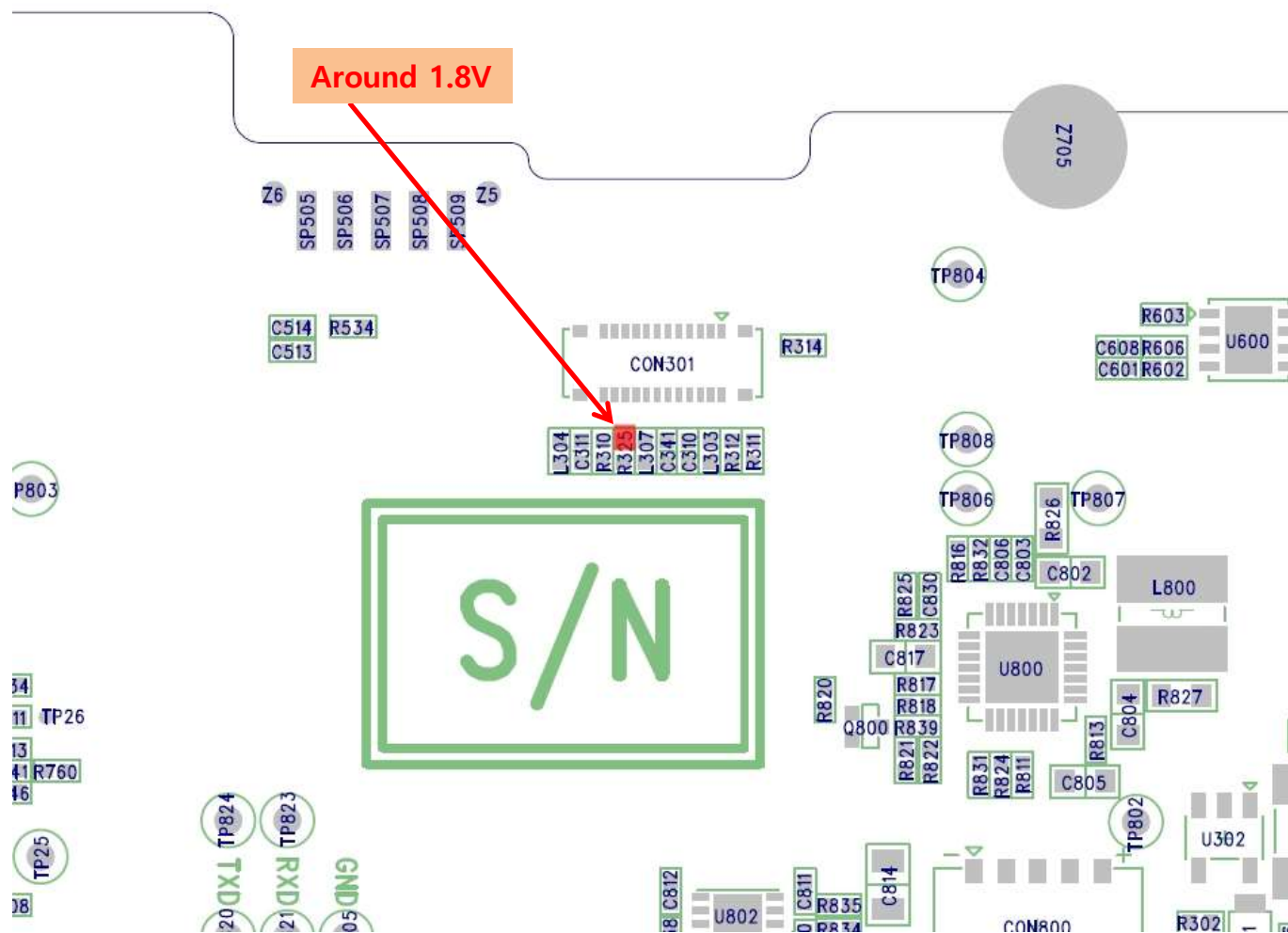
Camera power is
normal?

Check the power of the CAMERA, 1.8V(L303,
L307), 2.8V(L304).

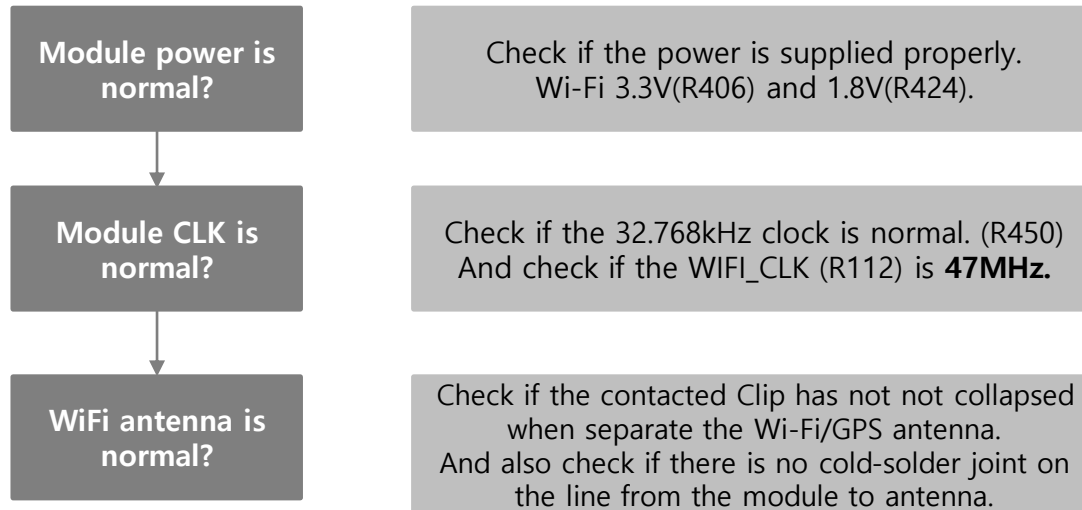


Camera RST is normal?

Check if the CAM RESETn (R310) is "High".



10. In case Wi-Fi isn't connected



Module power is normal?

Check if the power is supplied properly.
Wi-Fi 3.3V(R406) and 1.8V(R424).

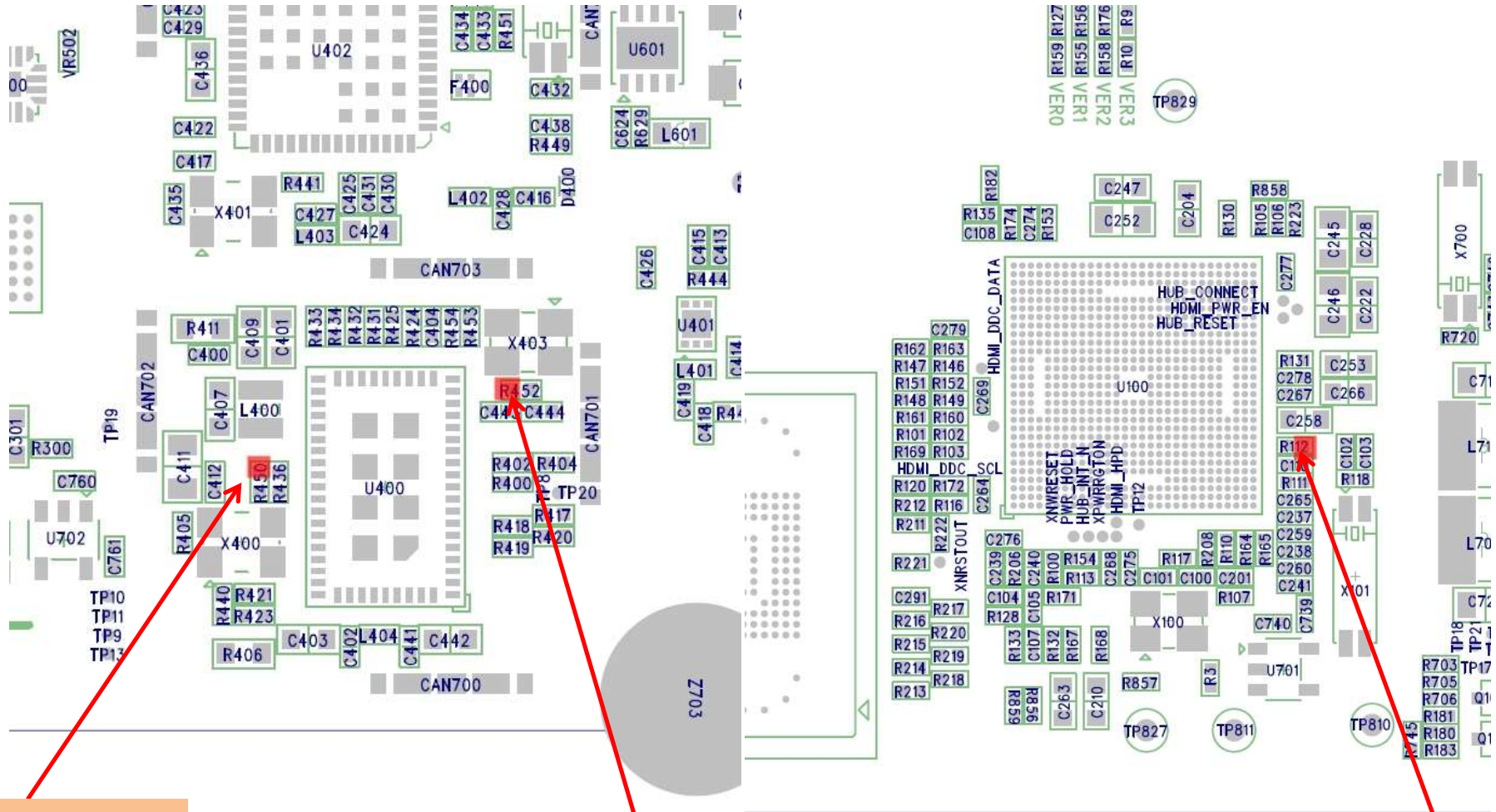
Around 1.8V

Around 3.3V



Module CLK is normal?

Check if the 32.768kHz clock is normal. (R450)
And check if the WIFI_CLK (R112) is 47MHz.

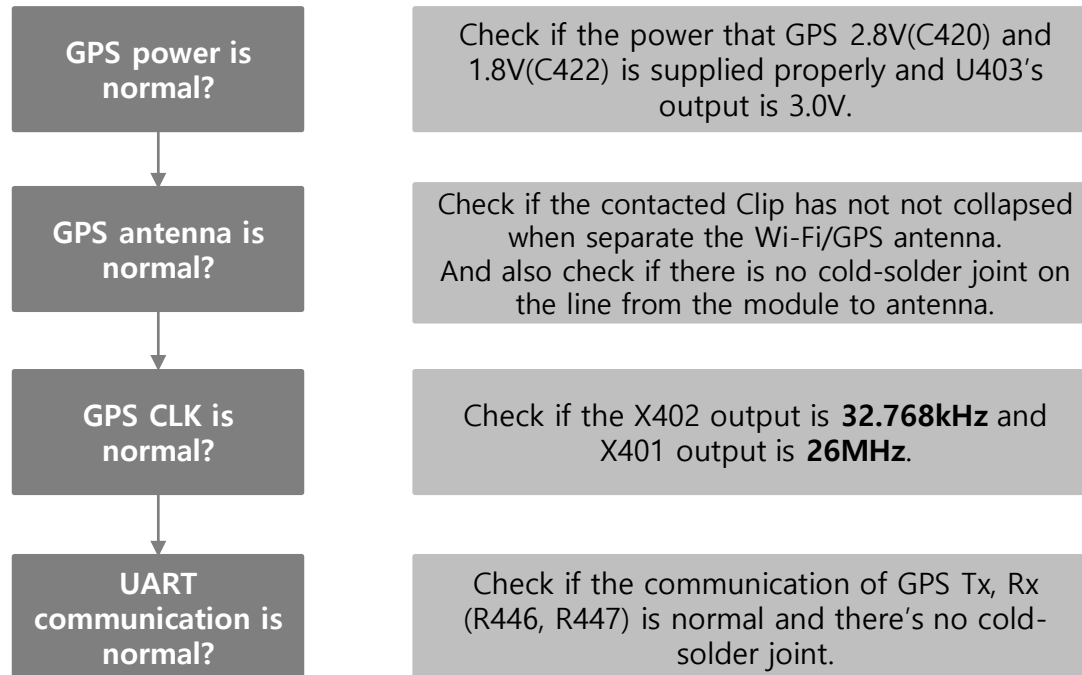


32.768kHz
From PMIC

37.4MHz
From Crystal (X403)

47MHz
From CPU

11. In case GPS doesn't work



GPS power is normal?

Check if the power that GPS 2.8V(C420) and 1.8V(C422) is supplied properly and U403's output is 3.0V.

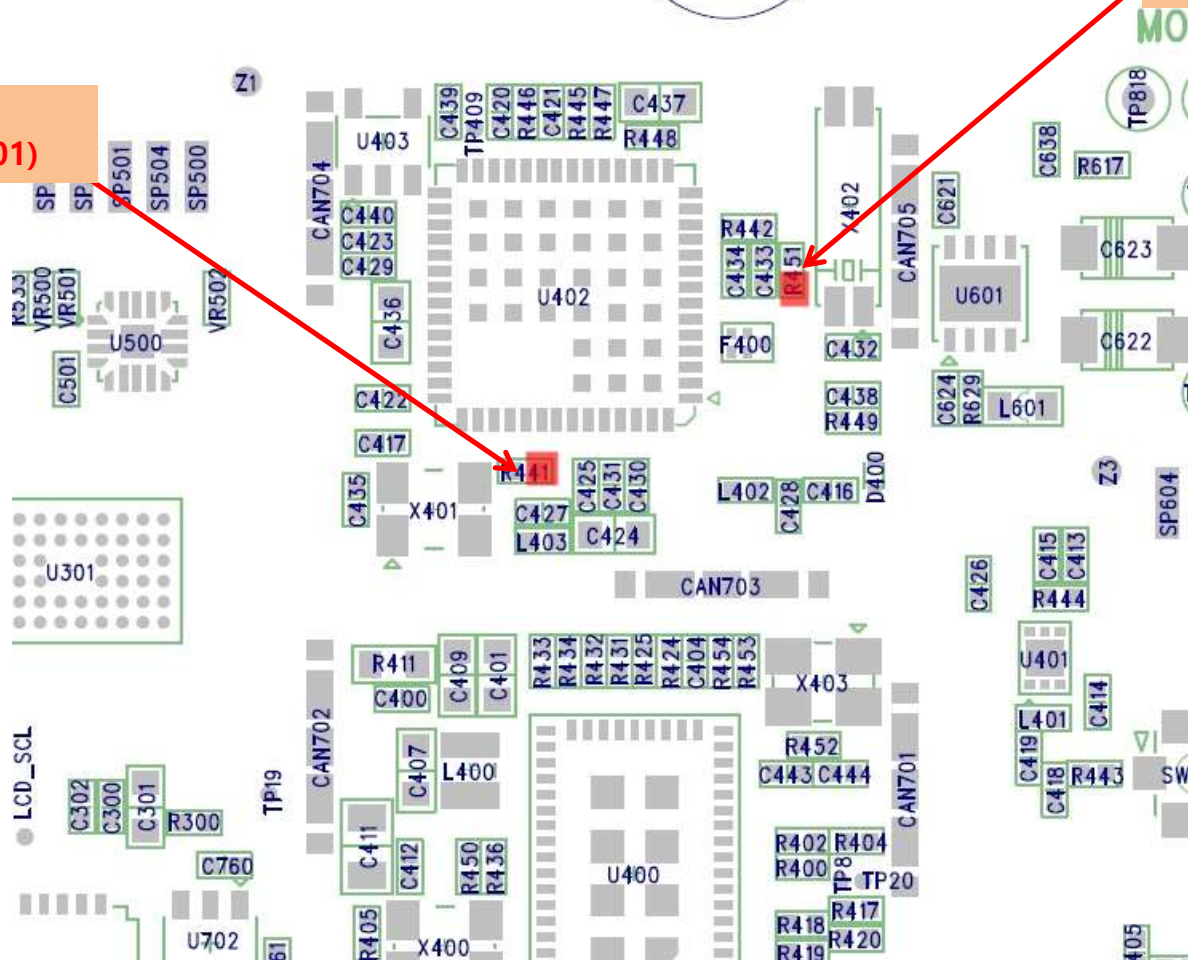


GPS CLK is
normal?

Check if the X402 output is **32.768kHz** and
X401 output is **26MHz**.

About 26MHz
From TCXO (X401)

About 32.768kHz
From PMIC



12. In case sensors don't work

